

AI and IoT: Supported Sixth Generation Sensing for Water Quality Assessment to Empower Sustainable Ecosystems

Suparna Das,[‡] Kamil Reza Khondakar,^{*,‡} Hirak Mazumdar,^{*,‡} Ajeet Kaushik,^{*} and Yogendra Kumar Mishra



Cite This: <https://doi.org/10.1021/acsestwater.4c00360>



Read Online

ACCESS |



Metrics & More



Article Recommendations

ABSTRACT: The integration of artificial intelligence (AI) and the Internet-of-Things (IoT) heralds the advent of sixth-generation sensing technologies, offering transformative potential for water quality assessment and the empowerment of sustainable ecosystems. This approach offers efficient monitoring through predictive analysis, risk assessment, and timely decision-making. However, this approach requires diverse expertise and requires connecting multiple dots. This paper explores the convergence of AI and IoT in developing advanced sensor networks capable of real-time monitoring and data analysis, providing comprehensive insights into water quality. AI algorithms can predict pollution events, optimize resource management, and enhance decision-making processes. IoT-enabled sensors provide extensive coverage and connectivity, facilitating continuous monitoring and immediate reporting of water conditions. This synergy ensures accurate detection of contaminants and supports proactive environmental management, aligning with global sustainability goals. Implementing AI and IoT in water quality assessment is crucial for maintaining healthy aquatic ecosystems, fostering biodiversity, and ensuring safe water resources for communities. The paper highlights the effectiveness and scalability of AI and IoT-supported sensing technologies, underscoring their critical role in a sustainable future.



1. INTRODUCTION

One of the most important resources in our world, water is necessary for all living things to survive and is fundamental to countless aspects of human society. Access to clean and safe water is non-negotiable whether it is required for consumption for drinking, agriculture, industrial processes, or ecological stability. However, its quality is threatened by pollution, climate change, and population growth. Water quality assessment is crucial for ecosystem sustainability and sustainable development. 70% of the earth's surface is water, and 60% of a person's body is water-based. According to the World Health Organization, clean and safe water is necessary for domestic use, industry, and health to prevent diseases like cholera, diarrhea, hepatitis, acne, and typhoid.¹ Lake pollution and sewage disposal are caused by population growth, urbanization, and modernity. Natural water sources are contaminated by rock weathering, soil leaching, mining operations, and other processes. Longer growing seasons and land use changes may increase fertilizer consumption, leading to eutrophication and biodiversity loss in rivers, lakes, and watercourses.

Over time, there has been a tremendous evolution in the development of technology for monitoring water quality, from simple manual procedures to sophisticated sensor systems. Each generation has led to advancements in automation, integration, sensitivity, and specificity, making it easier, faster,

and more accurate, improving water management and safety. The first-generation techniques, such as titration, colorimetry, gravimetry, and spectrophotometry, were manual and lab-based.^{2–4} The second-generation of portable on-site testing instruments, such as single-use, paper-based Microbial Fuel Cell (MFC) Sensor, Lab on a Smartphone (LOS), and portable total nitrogen analyzer was launched.^{5–8} The third-generation switched to an Integrated Sensor System with constant online monitoring.^{9,10} Intelligent sensors of the fourth-generation are integrated and include remote capabilities (multiparameter probes, wireless connectivity, Internet-based remote data access, and automated data processing).^{11,12} Online “big data” resources and smartphone-based surveys can be coordinated with distant sensors utilizing satellite technology and linked to the Internet-of-Things (IoT).^{13–15} Utilizing “big data” from meter fleets, high-resolution smart water meters and powerful data analytics enable the creation of an intelligent urban water management system.¹⁶ Drones are increasingly used for studying watersheds, using digital

Received: April 27, 2024

Revised: January 4, 2025

Accepted: January 8, 2025



photographs captured on board UAVs to generate three-dimensional point clouds for Structure from Motion (SfM), similar to LiDAR technology.¹⁷ The fifth-generation biosensors and microfluidics, including enzyme-based, DNA/RNA-based, immunosensors, and lab-on-a-chip devices, are crucial for the early detection of contaminants and changes in water quality, ensuring sustainable water resource use.^{18–22} The emergence of sensor technology along with LOC is being utilized for healthcare and environmental monitoring.^{23–27} 6G sensors are revolutionizing water quality monitoring by detecting temperature, dissolved oxygen, pH, and other parameters like heavy metals, pathogens, pesticides, and microplastics. These sensors offer extreme sensitivity, predictive capabilities, and adaptability, enabling accurate, effective, and user-friendly solutions for water resource management. Integrating with data analytics and remote communication systems enhances water assessment accessibility, empowering decision-makers to address environmental challenges and mitigate water security risks. There are several readily available sensor-based systems for tracking Sustainable Development Goal 6 (SDG 6) priority chemical arsenic exposure like the Palintest Digital Arsenic Test Kit.²⁸ SDG 6 aims for universal access to clean water and sanitation by 2030, focusing on health, well-being, and sustainable development through improved water quality, efficiency, and integrated water resource management.²⁹

The sixth generation sensing (6GS) is revolutionizing water quality assessment by providing real-time monitoring, predictive capabilities, and improved data analysis. This technology enhances water resource management and safeguards resources. The Internet of Things (IoT) is an interconnected network of devices that communicate and share data via the Internet, monitoring water quality aspects like temperature, turbidity, dissolved oxygen, pH, and contaminants.³⁰ AI is the application of computational methods, especially those modeled after the operations of the human intellect, to improve the monitoring, analysis, and control of water quality in natural settings.³¹ AI is transforming the way we analyze and make decisions about water quality, utilizing algorithms and models to analyze vast amounts of data, identify trends, and make informed decisions. It is also aiding in equipment selection, operation optimization, and solving complex problems that humans cannot handle.³² Polyaromatic hydrocarbons (PAHs) pose a significant environmental threat. Photocatalysis offers a promising solution for effective and long-lasting environmental management strategies, involving the detection, entrapment, and elimination of specific contaminants.³³

To analyze water quality for sustainable ecosystems, this article novels the application of 6GS [artificial intelligence (AI), IoT, and smart sensing technologies]. 6GS transforms the evaluation of water quality surpassing the capabilities of its predecessor, 5G by leveraging advanced materials, sophisticated algorithms, high-resolution, real-time monitoring and seamless connectivity. Cutting-edge sensors built into Internet of Things devices gather information enabling ultrasensitive detection of minute changes in temperature, turbidity, dissolved oxygen, pH, and nutrient levels, offering a thorough understanding of the condition of water bodies. 6GS uses AI and machine learning (ML) to detect sources of pollution, forecast changes in water quality, and send off early alarms for possible dangers by analyzing large data sets. Informed decision-making is facilitated by this, allowing for preemptive steps to protect aquatic ecosystems and provide sustainable

water supplies for coming generations. This discusses the challenges associated with traditional methods and how it is solved using AI and IoT technologies. The article also discusses the specific applications of AI and IoT in advancing smart sensing technologies, including various sensors that are in practice to gauge important factors. The article also examines case studies of AI-driven water quality monitoring systems deployed in various environmental settings, demonstrating their effectiveness in detecting pollution events, predicting water quality trends, and supporting prompt decision-making for environmental preservation and resource management. Finally, the article emphasizes the transformative potential of AI and IoT for revolutionizing water quality assessment practices and advancing sustainable ecosystem management.

2. WATER QUALITY ASSESSMENT: AN ESSENTIAL NEED

Clean water is crucial for maintaining healthy ecosystems, sustaining life, and supporting various aspects of human well-being. It is essential for personal hygiene, personal hygiene, and food preparation. Sustainable management of water resources and pollution prevention are key strategies to safeguard this precious resource for current and future generations. Recognizing the importance of clean water underscores the need for responsible water management, conservation efforts, and global initiatives to ensure its accessibility and protection.

Agriculture heavily relies on clean water for irrigation, sterilization, medical equipment, and patient care. Effective irrigation techniques require clean water. Food security and excellent agricultural output are associated with the consistent availability of clean water. Reduced growth and agricultural diseases might result from poor water quality. By minimizing the accumulation of pollutants and salts, which can lower the agricultural output and deteriorate soil quality, clean water helps preserve the health of the land. Livestock growth and health depend on clean water. The production of meat and dairy products may be impacted by infections and poor animal health caused by contaminated water.³⁴ Not only is clean water vital for ecosystem support, biodiversity preservation, and sustainable development, but it is also necessary for public health.

One of the most crucial requirements for growth that is both healthy and fruitful is having access to drinkable water. To avoid water-borne illnesses such as cholera, dysentery, and typhoid, it is suggested that everyone or every field have access to safe drinking water. Clean water is vital for food preparation, consumable drinking water, and general health management. Because polluted water can cause malnourishment and other health issues. Clean safe water is essential for good hygiene and sanitation habits, which stop the spread of infectious illnesses and improve public health.³⁵

Aquatic habitat preservation and biodiversity maintenance depend on clean water. Aquatic creatures can become extinct because of polluted water destroying their habitats. Diverse food webs, essential for preserving ecological balance and sustaining various species, including people, are supported by healthy aquatic habitats. Natural nutrient cycles, which are necessary for aquatic plant and animal development and reproduction, are supported by clean water. Controlling the quality of the water aids in limiting pollution, averting toxic algal blooms and hypoxic zones that can wipe out aquatic life. Clear bodies of water are important for recreation and tourism,

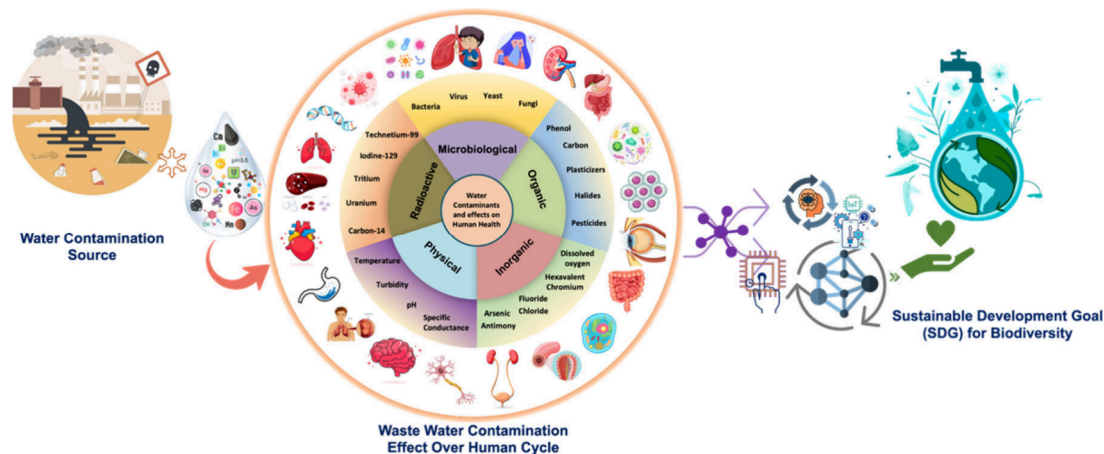


Figure 1. Illustration of water contamination sources (left) and contaminants in polluted water and related adverse effects on organs/health (center) and a roadmap where nanotechnology (right) supported 5th- and 6th-generation sensing for pollutant detection and suggested ways to get clean water for a sustainable ecosystem.

which boosts the local economy and well-being of the community. Millions people rely on fisheries and aquaculture for their food and livelihoods, which are supported by healthy aquatic ecosystems.³⁶ Water use for opulent baths and swimming pools has resulted from this, underscoring the need for clean water for both environmental and public health.

Factors like urbanization, industrialization, irrigated agriculture, and growing living standards have increased the need for fresh water. Clean water is crucial for maintaining healthy ecosystems, supporting plant life, and providing habitats for local wildlife. It enhances the visual appeal of recreational spaces like parks, fountains, and ponds, which provide serene environments for residents and visitors. Clean water features contribute to the overall beauty of urban landscapes, creating focal points for relaxation and social interaction. Well-maintained water bodies also improve air quality and reduce the urban heat island effect, enhancing the livability of city environments. Ensuring water availability and effective approaches for water cleaning are essential for ecological health, recreational enjoyment, and the visual and environmental quality of urban spaces.³⁷

2.1. Water Contaminants. Water contaminants are substances or pollutants found in water sources. They can impact the environment, and aquatic ecosystems (Figure 1). Table 1 represents some common water contaminants and their impacts.^{38–47}

2.2. Regulatory Standards for Water Quality. Regulatory standards are established by governments and international organizations to ensure water safety and protect aquatic environments. They set specific limits on contaminants and parameters in water, typically setting maximum contaminant levels (MCLs) for various water contaminants. Most nations maintain regulatory standards for drinking water quality monitoring to reduce waterborne illness risks. These rules cover surveillance monitoring of regulated and uncontrolled water sources and operational monitoring of water providers. Water regulatory standards vary significantly across regions due to environmental conditions, economic development, public health priorities, and governance structures. The United States Safe Drinking Water Act mandates water suppliers to test for and limit over 90 contaminants, while the EPA sets a maximum contaminant level for substances like lead at 15 ppb. The European Union's Drinking Water Directive outlines water

quality standards, monitoring requirements, and acceptable limits for contaminants.^{56,59}

3. OVERVIEW OF TRADITIONAL WATER QUALITY MONITORING METHODS

Conventional laboratory-based techniques, such as the first Generation, are still used to check water quality by physically gathering samples from water sources, analyzing them for pollution or microorganisms, and forecasting the general water quality using the Water Quality Index (WQI).⁶⁰ The WQI is a tool used to assess water quality. It uses traditional methods like field observations, grab sampling, in situ measurements, bacteriological analysis, macroinvertebrate sampling, sediment sampling, and laboratory analysis. These methods evaluate water bodies' color, clarity, odor, and presence of debris, algae, or other contaminants. Grab sampling, typically taken by hand, provides quick snapshots of water quality.⁶¹ When doing in situ measurements, no samples are taken; instead, the water quality parameters are measured right at the location of interest.⁶² Bacterial analysis tests water samples for bacteria, including total coliforms and *E. coli*, which indicate fecal pollution. Sediment sampling collects sediment samples from water bodies, revealing past contamination patterns and their impact on water quality. Studying sediment samples can provide insights into past pollution deposition patterns.⁶³ Traditional techniques have been used for years to analyze water properties in reservoirs, providing valuable information for understanding water quality, detecting contaminants, and ensuring its suitability for various applications. Water distribution networks and treatment facilities use specialized instruments to monitor water quality, identify impurities, and determine its suitability for human consumption.^{3,64}

Lab-based analytical techniques like chromatography, mass spectrometry, and spectrophotometry are crucial for water quality monitoring due to their high accuracy and specificity. These methods are particularly useful for detecting emerging contaminants like pharmaceutical residues, microplastics, and endocrine-disrupting chemicals. While biosensors are feasible for some applications, they have limitations in real-time monitoring. As biosensors develop, traditional methods will continue to play a critical role in accurate monitoring of emerging contaminants.

Table 1. Some of the Common Water Contaminants and Their Impacts and the Sensing Parameters to Implement Water Quality Monitoring Techniques

Contaminant Type	Contaminants	Sensing Parameters	Impacts	ref.
Inorganic Contaminants	Heavy metals (e.g., lead, mercury, arsenic)	Inductively coupled plasma mass spectrometry (ICP-MS), and atomic absorption spectrometry (AAS).	Heavy metals can accumulate in organisms, leading to toxicity and bioaccumulation.	48–50
Nutrients	nitrites and phosphates.	Chromatography	Excessive nitrates and phosphates may result in eutrophication, which depletes oxygen in aquatic habitats and causes algal blooms.	51
Organic Contaminants	Pesticides, herbicides, industrial chemicals (e.g., PCBs, dioxins), pharmaceuticals, volatile organic compounds (VOCs).	Gas chromatography–mass spectrometry (GC-MS), liquid chromatography–mass spectrometry (LC-MS), and fluorescence spectrometry.	The use of these substances can pose health risks to humans and wildlife, disrupt endocrine systems, and damage aquatic life, leading to population decline and an unbalanced ecology.	52, 53
Microbiological Contaminants	Pathogens- Bacteria (e.g., <i>Escherichia coli</i> or <i>E.coli</i> , Salmonella), viruses, protozoa (e.g., Giardia, Cryptosporidium).	Polymerase chain reaction (PCR), enzyme-linked immunosorbent assay (ELISA), and culture-based methods.	Waterborne diseases like cholera, dysentery, and gastroenteritis pose significant health risks, especially in areas with inadequate sanitation and hygiene practices.	54
Radiological Contaminants	Radon, uranium, cesium, strontium	Gamma spectrometry, liquid scintillation counting, α and β particle detectors.	Exposure to these substances can increase cancer risks and genetic mutations, leading to serious health effects such as organ damage and developmental anomalies.	55, 56
Sediment and Suspended Solids	Soil particles, silt, clay, organic matter	Turbidity meters, sediment traps, particle counters	Water quality is compromised by reduced light penetration, habitat smothering, and transporting pollutants, while aquatic ecosystems are disrupted by clogging fish gills and disrupting feeding behavior.	57
Microplastic	Microbeads, Synthetic fibres, plastics, Degraded fishing gears	Fourier-transform infrared (FTIR) spectroscopy and Raman spectroscopy, Pyrolysis-GC/MS (Gas Chromatography–Mass Spectrometry)	Marine organisms, including fish, plankton, often mistake microplastics for food, leading to physical harm and potential blockage of digestive tracts.	58

Conventional methods for water quality evaluations have limitations like labor-intensive sampling, time-consuming analysis, limited spatial coverage, inability to capture real-time data, high costs, and limited parameter measurement. They struggle to identify pollution sources, lack early warning, seasonal variability, and are incompatible with long-term monitoring. The outdated approach is expensive, time-consuming, and impractical, potentially leading to delays and human errors. To streamline the WQI calculation process, an alternate technique is needed to improve precision, effectiveness, and thoroughness.

Geospatial big data is bringing revolutionary changes in society and research with the ability to benefit public health, the economy, and other advancements like data-driven and community science. Leveraging geographic big data for applications in both fields is largely dependent on AI, particularly its branches of ML, deep learning (DL), and computer vision (CV). The single biggest source of geographic big data is remote sensing, which has shown significant advancements in both temporal and spatial resolution. In the meantime, recent developments in DL and CV have greatly improved remote sensing and geoscience research.⁶⁵ AI technologies can automatically identify and monitor large-scale water bodies and their quality, assisting in informed decision-making and prompt solutions to water quality issues. AI algorithms can analyze this data to identify trends and abnormalities and use machine learning models to forecast future changes in water quality. This improves analysis and proactive management, while also enhancing water treatment plant operations by lowering costs and increasing efficiency through real-time adjustments based on sensor data.

4. INTRODUCTION TO MODERN SENSING TECHNOLOGIES

4.1. Biosensors. The ability of biosensors to detect biological effects, their portability, and the ability to enable on-site monitoring have made them an efficient instrument for the rapid and focused observation of a variety of analytes in context.^{66–69} In 1962, biosensors were created to measure Biochemical Oxygen Demand (BOD) in the nascent phases of the technology. Chee et al.⁷⁰ has created a yeast-based optical fiber biosensor that can detect low BOD levels in river waters in 15 min. A few biosensors have been fabricated to track minute concentrations of nitrogen molecules in water. Portable paper-based biosensors were developed for the onsite detection of pesticides, heavy metals, and bacteria,⁷¹ enzymatic luminous biosensors, and nanoelectrochemical biosensors for environmental toxicants (heavy metals and insecticides)^{72,73} and waterborne pathogens.⁷⁴ Figure 2 shows the presence of various contaminants in the environment and showcases the process of monitoring and detecting these environmental contaminants using biosensors for subsequent data analysis, contributing to our understanding and mitigation of environmental issues.

Microbial fuel cell (MFC)—based biosensors offer self-powered devices for chemical oxygen demand, dissolved oxygen, and BOD early warning water quality monitoring. However, they have a lengthy start-up time and poor power output, and their performance depends on environmental factors like pH and temperature. Traditional biosensors require costly and time-consuming processes, while MFC sensors are less useful for urgent monitoring due to their dependability on environmental factors.^{75–77} MFC sensors are cost-effective,

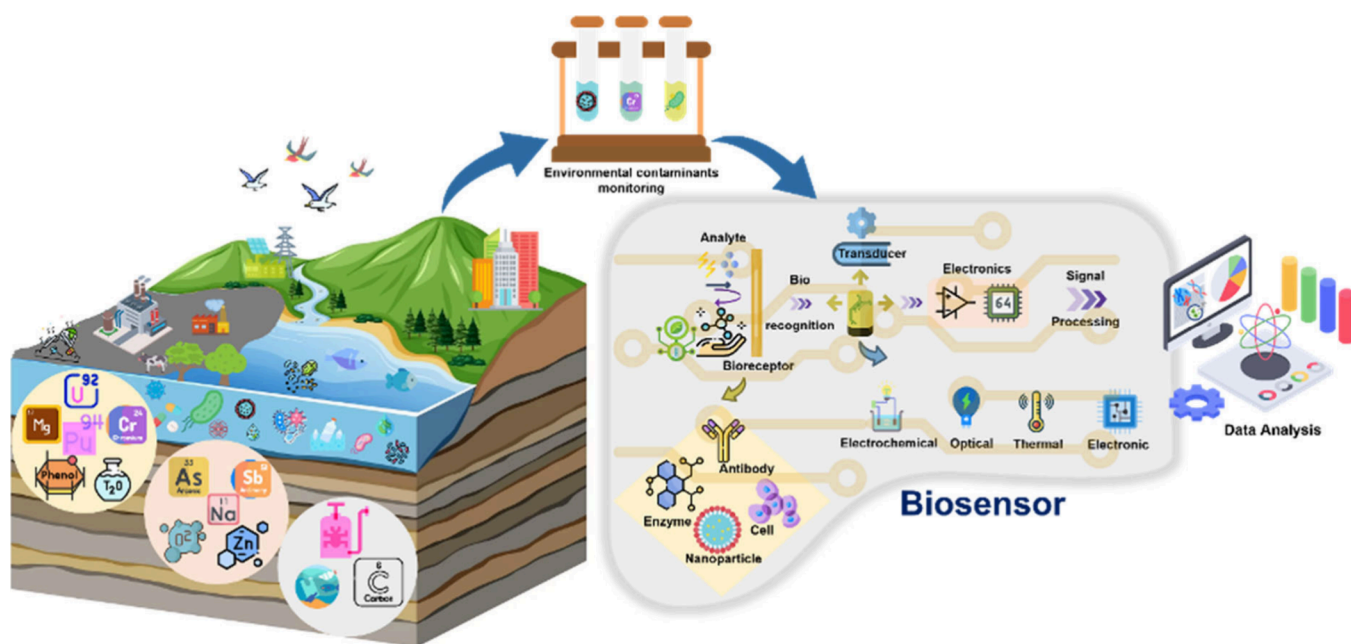


Figure 2. Different types of pollution present in the environment and the monitoring/detection of environmental contaminants through biosensors for data analysis.

biodegradable, compact, and portable, making them ideal for real-time monitoring of water and air quality on-site.^{78,79} Interestingly, a self-powered, autonomous MFC biosensor for online water quality monitoring based on signal frequency has been reported by Pasternak et al.,⁸⁰ to detect urine contamination in water. Chouler et al.⁸¹ investigated the use of microalgae in photosynthetic microbial fuel cells (photoMFCs) to create a sensitive, portable, and affordable bioelectrochemical sensor for on-site monitoring of contaminants in algal pond water. The photoMFC sensor has demonstrated superior effectiveness in detecting on-site pollutants like formaldehyde and herbicides, as well as pharmaceuticals. It uses DNA aptamer and quantum dots^{82,83} to detect aquatic toxins produced by freshwater and marine microorganisms. This technology has been used to detect water-borne bacterial infections, agricultural poisons, heavy metals, and other environmental pollutants.⁸⁴

Microfluidic biosensors manipulate and analyze water samples using microscopic channels and chambers to assess water quality characteristics, including bacteria, viruses, ions, nitrate, and toxins.^{85,86} Bikman et al.⁸⁷ recently reported a demand in the aquatic industry by using an innovative portable biosensor system for the rapid detection of freshwater cyanobacterial algal bloom toxins. Considering recent advancements,⁸⁸ the design and development of accessible smart metal oxide nanostructures (MO-NSs) for water quality management have been discussed. Given the difficulty of eliminating pharmaceutical contaminants from wastewater, Letsoalo et al.⁸⁹ described a wide range of newly discovered pharmaceutical contaminants, their biotoxicity, potential health risks, methods for detecting them, and remediation strategies that use a nanosystem with adjustable properties that are managed through the use of green practices. Furthermore, Kumar et al.⁹⁰ designed State-of-the-art technology-assisted smart sensors like point-of-care (POC) technologies with advanced lateral flow devices, microfluidic systems, and paper-based assays to detect Waterborne Pathogens causing disease. Wang et al.⁹¹ created a

self-powered biosensing device that uses the vertical contact-separation triboelectric nanogenerator (TENG) to identify *Staphylococcus aureus* (*S.aureus*) in solution, a Gram-positive bacterium, by analyzing the biosensor's recorded voltage fluctuation. Table 2 highlights the diverse capabilities and limitations of biosensors, including their detection mechanisms and specific uses in assessing water pollutants, pathogens, and other quality indicators.

4.2. Smart Sensing Technologies: AI and IoT. Modern sensing technologies are advancing water quality monitoring, offering real-time measurement capabilities. Researchers are developing Raspberry Pi-based hardware platforms for water monitoring systems. Sensors are being integrated into drones and unmanned vehicles for remote water quality information.¹¹² Remote sensing techniques, including satellite and aerial imagery, can also provide crucial data on water quality. This advancement in water monitoring technology is crucial for addressing water quality issues.¹¹³

Hyperspectral imaging captures high-resolution images over the electromagnetic spectrum, offering real-time monitoring capabilities from satellite or aircraft platforms.^{114,115} An online UV-vis spectrophotometer was proposed for process management and drinking water quality monitoring, providing continuous and accurate assessments with shorter reaction times and no sample pretreatments or reagents. Surface-enhanced Raman Scattering (SERS) is emerging as a powerful method for fingerprint information for trace compounds, based on ultrasensitive vibrational spectroscopy on the surface and water interface. Cancui et al.¹¹⁶ suggest SERS as a new modern bacteria detection method. Su et al.¹¹⁷ suggested that it has mostly been used for the identification and detection of organic pollutants in water. Furthermore, Du et al.¹¹⁸ developed a method using fluorescence intensity ratio and intensity to determine bacteria amounts, enabling efficient and quick detection using laser-induced fluorescence (LIF) spectroscopic technology.

Table 2. Comparison of Biosensors for Water Quality Monitoring

Type of Biosensor	Principle	Abilities	Drawbacks	Applications	ref.
Electrochemical Biosensors	Detect changes in electrical properties (e.g., current, voltage) due to biochemical reactions.	High sensitivity, rapid response, and ability to detect heavy metals and organic pollutants.	Susceptible to fouling, limited lifetime, and can require frequent calibration.	Monitoring Gram-positive bacteria, heavy metals, pathogens, pesticides, and nitrates in water	74, 91, 92
Optical Biosensors	Use light absorption, fluorescence, or refractive index changes caused by analyte interaction.	High specificity, real-time analysis, and ability to monitor multiple analytes simultaneously.	Expensive setup, sensitive to environmental conditions, and sometimes bulky	Detection of pathogens, BOD, organic pollutants, and Cyanobacterial Algal Bloom toxins.	93, 94
Fiber-Optic Biosensors (Zinc oxide nanorods, <i>Chloridia vulgaris</i>)	Utilize light transmitted through optical fibers to detect analytes.	Highly sensitive, flexible, and suitable for remote or in situ monitoring.	Fragile, requires complex alignment, and can be costly.	Monitoring pesticides, <i>E.coli</i> , toxic compounds, pollutants, heavy metals, and pathogens in water.	95–97
Enzyme-Based Biosensors (Hand-held Enzymatic Luminescent Biosensor)	Detect analytes through specific enzyme-analyte interactions producing measurable signals.	High specificity and fast response time.	Limited stability due to enzyme degradation and sensitivity to temperature and pH.	Monitoring heavy metals ions, BOD, ammonia, toxic elements, catechol and pesticides.	98–100
Microbial Biosensors	Use whole microbial fuel cell (MFC) to detect analytes through metabolic activity.	Versatile, cost-effective, and capable of detecting a wide range of organic and inorganic compounds.	Slower response times and can be affected by environmental factors (e.g., toxins or temperature).	Monitoring BOD, toxicants, organic pollutants, heavy metals, and wastewater treatment efficiency.	78, 79, 81, 101
Aptamer-Based Biosensors	Utilize nucleic acid hybridization or DNA interactions for analyte detection.	Extremely high specificity, capable of detecting genetic material of pathogens.	Complex preparation, expensive, and requires expertise to operate.	Pathogen identification and monitoring genetic contaminants. Detect Aquatic Phycotoxins and Cyanotoxins. Detect Low Molecular Weight Pollutants	84, 102–107
Photonic Immunosensors	Use antigen–antibody interactions to detect specific analytes.	High specificity and can detect very low concentrations of pathogens or toxins.	Limited reusability, high cost, and sensitivity to environmental conditions	The detection of pathogens, toxins, and endocrine disruptors, particularly <i>E. coli</i> , is crucial in bathing and drinking water.	108–111

Paz et al.¹¹⁹ developed an open-source and end-to-end wireless platform for real-time monitoring of various parameters like pH, electrical conductivity, temperature, and pressure in drinking water systems. Figure 3(a, b) shows a small sensor chip with a microcontroller with a data collection facility for analysis.¹²⁰ Alam et al.⁶² fabricated an electrochemical sensing system that uses a reading circuit, a potentiostat, and an Android app to simultaneously measure pH, free chlorine, and heavy metals in water samples (Figure 3c shows the prototype device with data analysis of water samples). The advantages of integrating ML, Raman spectroscopy, and microfluidics are demonstrated in Figure 3(d–f), which aims to increase the precision and effectiveness of microplastic characterization in marine settings.¹²¹ An affordable electrochemical PCB sensor can detect viruses in environmental samples, offering a practical and economical alternative to immobilized oligonucleotides electrodes or DNA sensing substrates, which require low-temperature storage.¹²² Here, three phases comprised the DNA synthesis reaction: reverse transcription at 46 °C for 20 min, reverse transcriptase inactivation at 94 °C for 1 min and priming at 25 °C for 5 min (Figure 3g). The development of a fluorescence-based biosensor for selective and sensitive detection of Pb²⁺ ions using a DNazyme system is expected to offer new insights into the quick, accurate, and sensitive identification of significant pollutants for food safety.¹²³

Important water-based applications, such as river-level monitoring, adsorption, membrane filtration, chlorination, monitoring of the water quality index, modeling of the water quality parameters, and automation or monitoring of aquaponics and hydroponics, have been combined with AI or ML.¹²⁴ A real-time data on water quality parameters is gathered by smart water quality monitoring networks using a range of sensors.^{125–129} Generated water quality data is analyzed using AI-powered water quality monitoring systems to know patterns and trends. AI methods based on image analysis are effectively used for water quality monitoring and water pollutant detection.¹³⁰ Internet-enabled systems with flow, pressure, and water quality. Characteristic sensors are common examples of IoT in water applications. The effective use of IoT has contributed to lowering yearly water consumption, improving it, and real-time monitoring of water usage.¹³¹

A sustainable ecosystem is a resilient, self-sustaining environment that provides habitat and resources for various organisms while minimizing environmental impact. Integrating AI with smart industry and healthcare can lead to numerous benefits, such as reducing waste, increasing efficiency, and minimizing environmental impact. This aligns with sustainability goals by promoting responsible resource management, cleaner energy use, and the well-being of individuals and the planet. The Industrial Internet of Things (IIoT) is industrializing, producing a significant amount of sensing data from various devices (Figure 4). Federated learning (FL) needs to be integrated with smart city applications to protect sensitive information and maintain privacy. This also promotes telehealth and telemedicine, improving accessibility to healthcare services.^{132–136}

AI-powered IoT devices can collect real-time data on various attributes, including machine performance, environmental conditions, and critical indications.¹³⁷ This synergy between AI and IoT in environmental monitoring can transform industries by raising standards of care, cutting costs, and

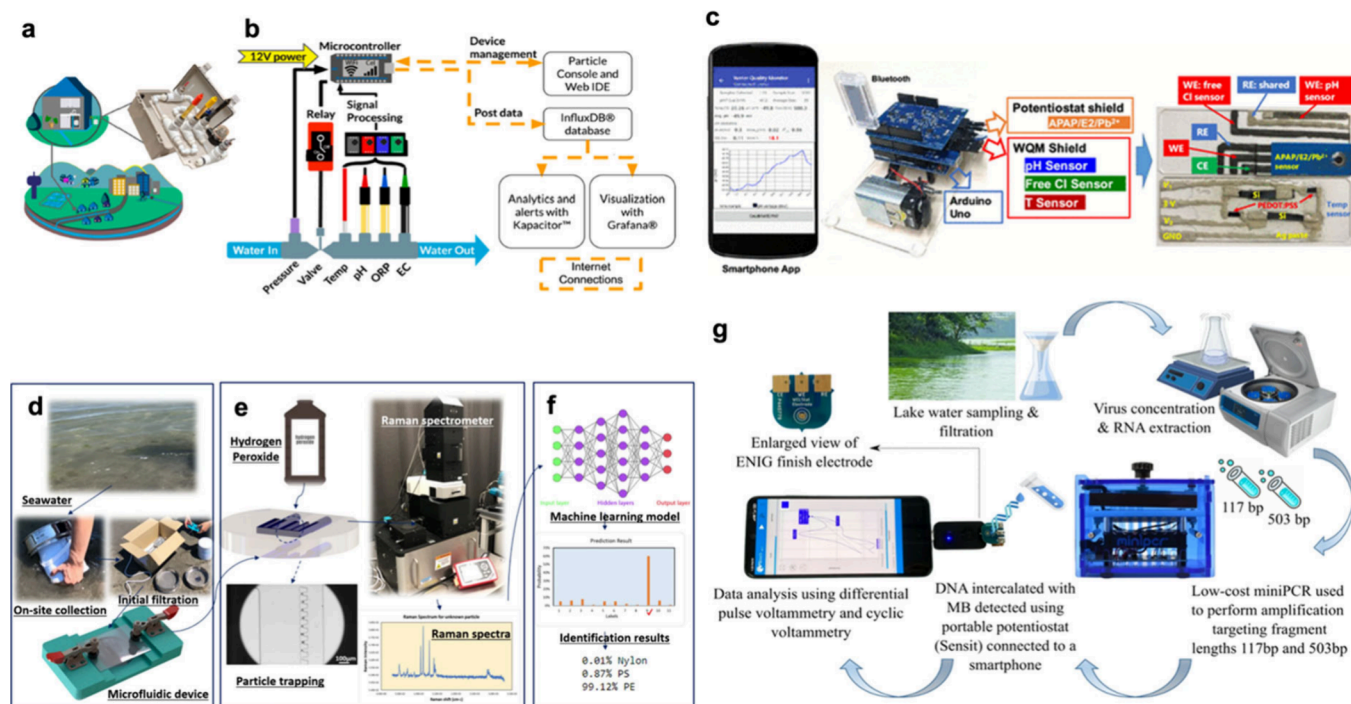


Figure 3. (a, b) A wireless sensor system for measuring drinking water quality using a microcontroller. Reproduced from ref 120. Copyright 2022 American Chemical Society. (c) An affordable array of electrochemical sensors for monitoring water quality in situ. Reproduced from ref 62. Copyright 2020 American Chemical Society. (d–f) A microfluidic device is used on-site to collect seawater samples, initially filtered with a 45 μm filter; after laboratory transport, a 30% hydrogen peroxide solution processes the samples for 16 h, followed by Raman spectra acquisition and ML-based identification of marine microplastics. Reprinted with permission under a Creative Commons CC-BY 4.0 from ref 121. Copyright 2023 Nature. (g) A mobile electrochemical sensor to identify viruses in lake water. Reprinted with permission under a Creative Commons CC-BY 4.0 from ref 122. Copyright 2022 Scientific Reports.

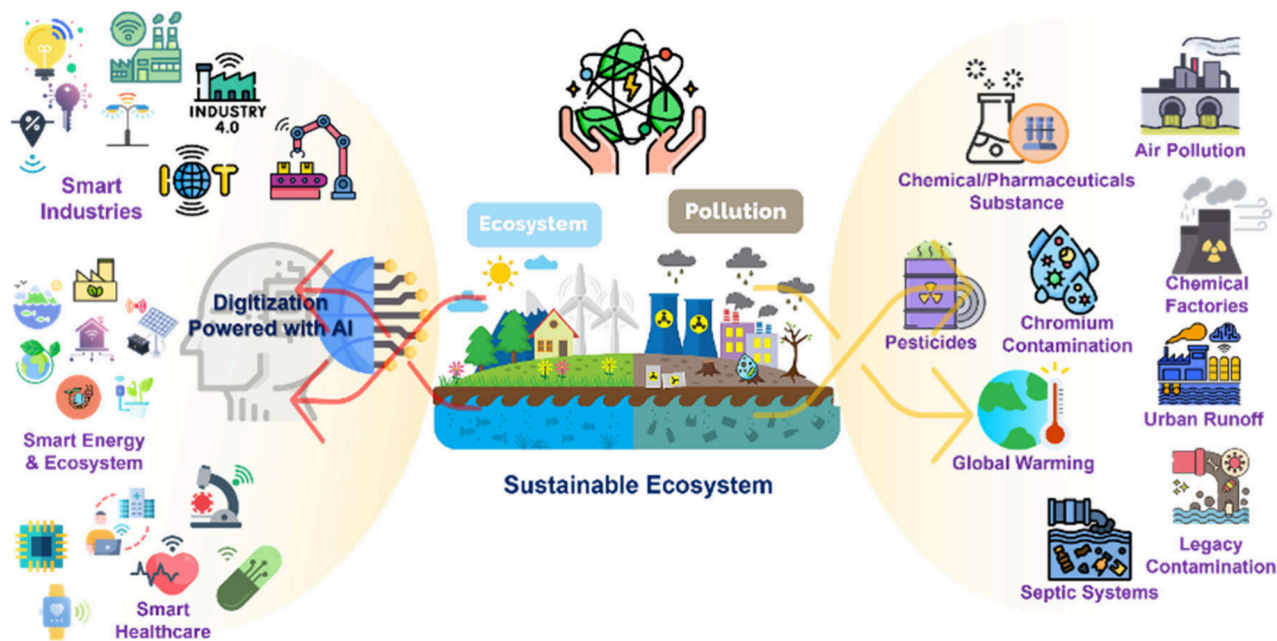


Figure 4. An ideal sustainable ecosystem integrating AI with smart industry and smart healthcare.

enhancing our understanding of environmental conditions.^{138–141} IoT sensors can measure temperature, humidity, pollution levels, and air and water quality, which can be analyzed using AI systems to identify environmental risks, forecast trends, and provide early warnings of natural disasters or environmental crises.^{142,143} IoT gadgets, such as remote

monitoring systems and water sensors, can collect data continuously on various water quality parameters. AI can process this data to identify abnormalities or contaminants, enabling remote monitoring and early detection of health risks and preventing widespread waterborne diseases.¹⁴⁴ AI can optimize water treatment processes by analyzing data from IoT

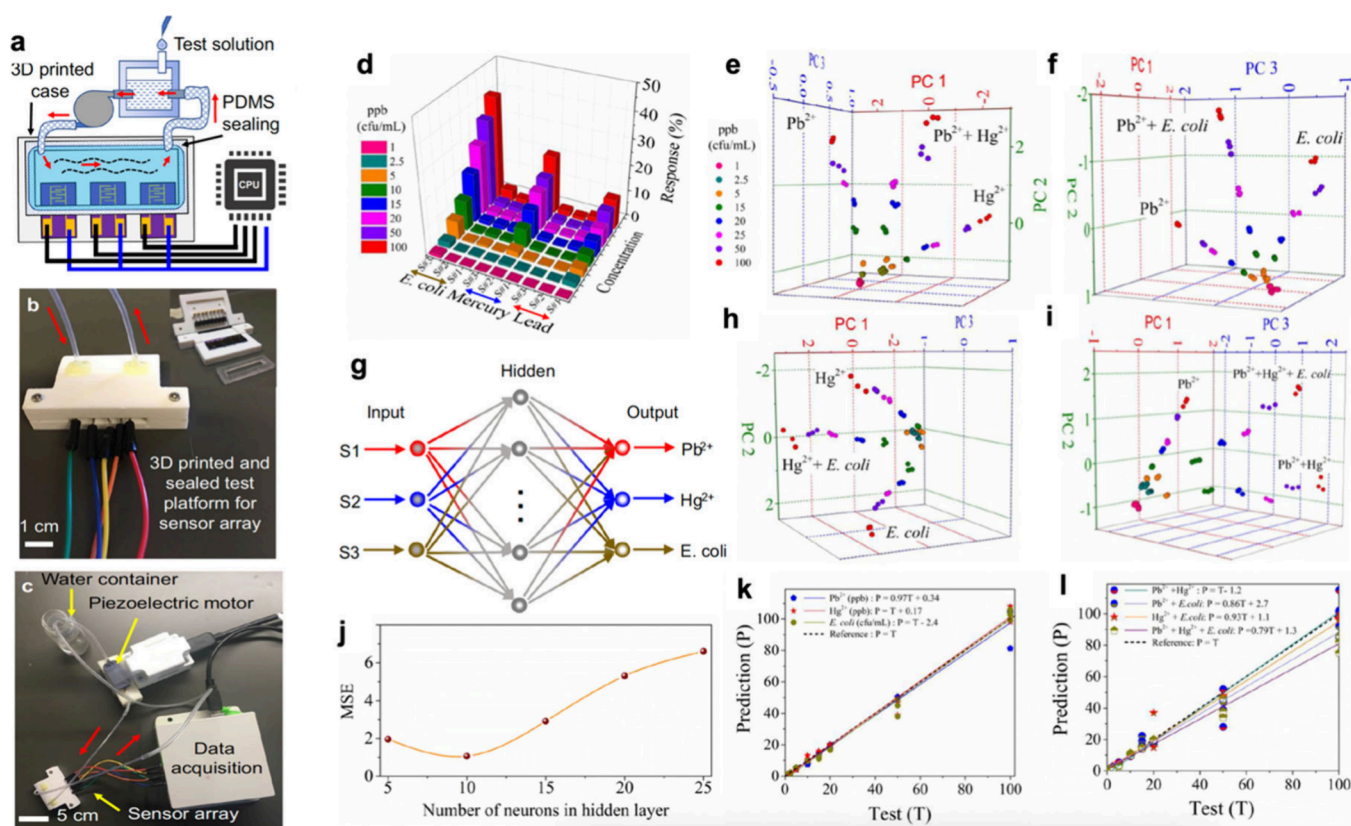


Figure 5. Measurements of a GFET sensor array in a flowing water system. (a) Schematic experimental setup for continuous sensing. The sensor array housed in the chamber is sealed with a polydimethylsiloxane (PDMS) mold from the source-drain contact. Water is drawn by the piezoelectric motor into the sensor chamber and then continuously flows back to the external container to mimic the water flow. (b) Photograph of a 3D-printed sealed chamber with the sensors embedded inside. Inset is the cell interior. (c) Photograph of a piezoelectric micromotor and an external water container connected to the sealed sensor chamber. (d) Summary of responses from Sensors 1–3 for Pb^{2+} , Hg^{2+} , and *E. coli*. (e–h) Principal component analysis (PCA) plots for the classifications of Pb^{2+} , Hg^{2+} , *E. coli*, and their mixtures. (i, j) Schematic of a two-layer ANN used for training and mean squared error (MSE) values for various numbers of neurons in the hidden layers. (k, l) Plots of the test and the predicted concentrations for Pb^{2+} , Hg^{2+} , and *E. coli* (ppb or cfu/mL) and their mixtures from the trained ANN model. Here, the dashed line ($Y = T$) represents the perfect match between the test and the predicted concentrations for unknown samples while the solid line represents the fitting of the target concentrations predicted from the trained ANN model. Reprinted with permission under a Creative Commons CC-BY 4.0 from ref 154. Copyright 2023 Nature.

sensors in treatment plants, adjusting chemical dosages, filtration rates, and other parameters in real-time to maintain water quality standards efficiently and reduce operational costs. Water quality monitoring is becoming more accurate, efficient, and economical due to smart sensing technologies that combine AI and IoT, leading to better water quality outcomes for the environment.¹⁴⁵ AI-powered smart sensors can be more accurate and sensitive than traditional sensors, compensating for sensor drift and other sources of error. The use of these technologies can also decrease the price of monitoring water quality by automating tasks such as sample collection and analysis.¹⁴⁶

The feasibility of evaluating water quality with a focus on agricultural water usage using a virtual sensing system based on physical sensor techniques.¹⁴⁷ Yi et al.¹⁴⁸ Yi et al. have developed an AI-biosensing framework for automated pathogen detection in water samples, including agricultural and liquid food. The system uses a dynamic learning (DL) system to identify and measure target bacteria based on their interactions with bacteriophages. Researchers are also experimenting with intelligent analysis using smartphones and AI algorithms to provide portable detection capabilities.¹⁴⁹ Wastewater-based epidemiology (WBE), a type of wastewater

monitoring, offers an additional clinical testing approach and may be able to track viral epidemics like COVID-19. AI-related ML and DL technologies are now being developed as methods for utilizing global data to identify virus epidemics.^{150–152}

5. AI AND IOT APPLICATIONS IN WATER QUALITY ASSESSMENT

ML is a powerful tool for obtaining information from databases, making it ideal for applications involving virtual sensors. From 2019 to 2023, several ML algorithms were used for water quality monitoring systems, with Artificial Neural Network (ANN) modeling being the most popular.¹⁵³ Over the past decade, anomaly based techniques for detecting water pollution using AI have emerged, including wet transfer, ML, and impedance and noise measurements. These techniques enable efficient diagnosis of malfunctioning devices and the scalable nanofabrication of graphene-based field-effect transistor (GFET) sensor arrays, which can detect *E. coli* bacteria and heavy metal ions in real-time in flowing tap water. As seen in Figure 5 (a–c), the GFET sensor array, including three separate sensor chips, is contained in a 3D-printed closed-loop chamber that is driven by a piezoelectric pump. This pump allows water to flow between the chamber and the external

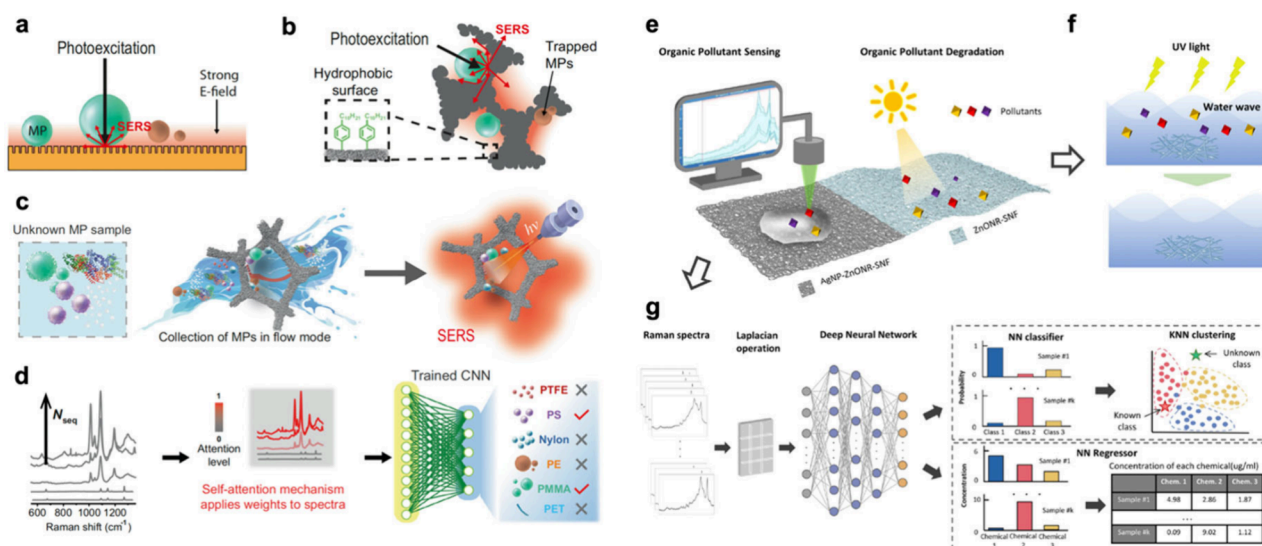


Figure 6. (a–d) Porous plasmonic substrates with a self-attention-based neural network enhance SERS-based detection of MPs in water contamination by trapping MPs, enhancing SERS signals, and using SpecATNet to focus on relevant spectra for accurate identification in complex mixtures. Reprinted with permission under a Creative Commons CC-BY 4.0 from ref 164. Copyright 2024 Nature. (e) Components of the AgNP-ZnONR-SNF system for organic pollutant detection and degradation. (f) The diagrammatic sketch of the photo- and piezo-catalytic degradation process. (g) The schematic demonstration of the ML models. Reprinted with permission under a Creative Commons CC-BY 4.0 from ref 165. Copyright 2024 Nature.

container through the inlet and output vessels. Figure 5 (d–l) illustrates the ANN modeling for classifying and quantifying various mixed ionic and bacterial species in flowing water.¹⁵⁴

An ANN and a support vector machine (SVM) are frequently used to classify water quality values into abnormal categories.¹⁵⁵ Enhanced AI algorithms, such as long short-term memory (LSTM) DL techniques and nonlinear autoregressive neural networks (NARNET), have been developed for Water Quality Indicator (WQI) prediction. For WQC forecasting, SVM, k-nearest neighbor (K-NN), and Naive Bayes are employed. Other algorithms, such as partial least-squares regression (PLS), support vector regression (SVR), cubist regression (CUB), and quantile regression neural network (QRNN), have been chosen due to their effectiveness in wastewater treatment predictions.^{156,157}

Soft sensor modeling uses a variety of ML algorithms and statistical learning approaches, particularly in the water quality sector.^{158,159} Djerioui et al.¹⁶⁰ evaluated the accuracy and reaction time of the Extreme Learning Machine (ELM) and SVM using a soft sensor based on chlorine and concluded that ELM outperformed SVM. Since Multiple Linear Regression (MLR) is the simplest approach, many researchers utilize it to predict water quality. To predict the concentrations of fecal indicator bacteria on beaches in rivers, lakes, and marine settings, MLR is a straightforward and interpretable approach.^{161,162} The Random Forest (RF) algorithm is a promising method for forecasting water quality indicators like total nitrogen (TN), total phosphorus (TP), and turbidity (TUR) in various research fields. It uses climate and watershed variables to predict water quality indicators in different landscapes, using techniques like stepwise regression, Least Absolute Shrinkage and Selection Operator, and genetic algorithm. The RF algorithm is easier to train and robust to model overfitting. Partial dependence plots reveal complex relationships between individual predictors and water quality indicators. The median values of total nitrogen and total phosphorus in streams increase significantly when urban and

agricultural land percentages exceed 40% and 43% of the watershed area, respectively. The SVM approach is also used to incorporate soft sensor principles in water quality assessment for parameters like COD, BOD, Chlorine, and TP. The Instance-Based Learner (IBK) algorithm is an ML-based soft sensor model that offers advantages like cost-effectiveness, efficiency, and reasonable accuracy. The KNN technique has demonstrated efficacy in COD prediction and assessment concerning response time and other performance parameters.¹⁶³ Using a self-attention mechanism, Guselnicova et al.¹⁶⁴ developed a neural network (NN) algorithm called SpecAT-Net to resolve complex dependencies and patterns in SERS data and identify six common types of MPs: nylon, polystyrene, polyethylene, poly(methyl methacrylate), polytetrafluoroethylene, and polyethylene terephthalate. This allowed the monitoring of complex samples with multiple MPs and the demultiplexing of numerous overlapping patterns (Figure 6a–d). The schematic diagram of the pollutant degradation, detection, and ML algorithm-assisted data interpretation process of the dual functional device by incorporating SNF thin film with ZnO nanorods (ZnONRs) and silver nanoparticles (AgNPs) through SERS is shown in Figure 6(e–g).¹⁶⁵

In every domain including civic, industrial, and agricultural, groundwater is an essential resource for living beings. Prediction of groundwater contamination made of different chemical constituents is essential for groundwater resource management, policy formation, and planning. To estimate groundwater quality, Sudhakar et al.¹⁶⁶ employed a DL-based model and compared it with three other ML models: RF, XGBoost, and ANN. The most accurate and optimal prediction model, according to the results, is the DL model. Haggerty et al.¹⁶⁷ discovered in another study that although neural networks have been less popular in recent years due to the advent of more sophisticated or accurate methods like DL or unsupervised algorithms, they are still the most often used ML model for groundwater quality (GWQ) modeling. Using

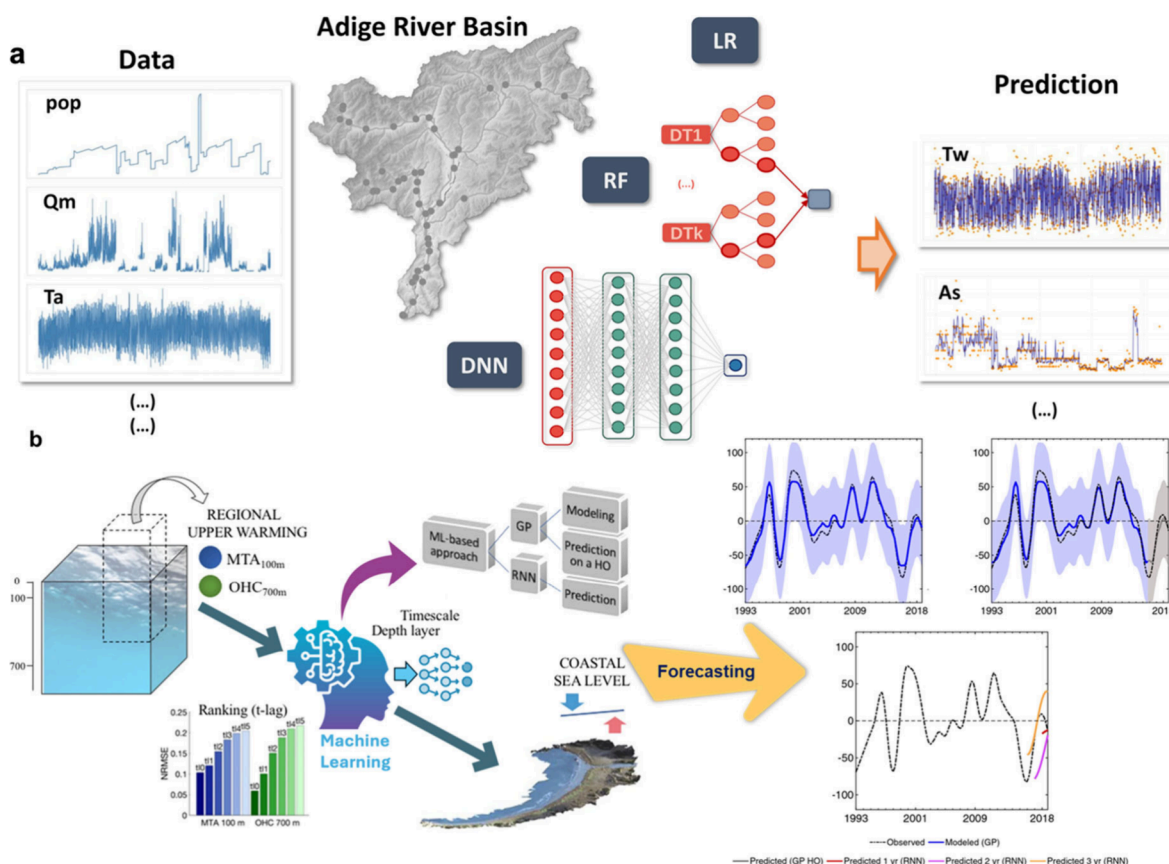


Figure 7. (a) A comparative analysis of the performance of Random Forest, Deep Feed-forward Neural Network, and Linear Regression models in predicting river water quality parameters such as temperature, dissolved oxygen, arsenic, sulfate, and chloride concentrations. Reproduced with permission from ref 168. Copyright 2022 Science of the Total Environment. (b) A combination of the nonlinear GP and RNN methods to assess coastal sea level variability in 26 coastal regions surrounding the Pacific, Indian, and Atlantic basins across a range of time scales. Reprinted with permission under a Creative Commons CC-BY 4.0 from ref 170. Copyright 2021 Scientific Reports.

RF and Deep feed-forward Neural Network (DFNN), a catchment-scale model of water temperature, dissolved oxygen, arsenic, sulfate, and chloride concentrations, as well as electrical conductivity, was created in an Alpine watershed¹⁶⁸ (Figure 7a). Tianan et al.¹⁶⁹ reported ML models for marine water management. These models have demonstrated their capacity to offer valuable insights and implications for comprehending the mechanisms and progression of Harmful Algal Bloom (HAB) outbreaks in coastal areas. This work provided a prediction to enhance the accuracy of water quality, thereby helping to achieve more efficient management of coastal hydro-environments. In this context, Nieves et al.'s¹⁷⁰ ML approach used ocean temperature estimates to model coastal sea level variability and uncertainty. It estimated near-future regional sea levels and performed well in coastal areas influenced by internal climate variability. The models are applicable globally for evaluating rising and falling sea levels. (Figure 7b)

Industries frequently discharge their wastewater, which is full of hazardous emerging contaminants (ECs) and toxic heavy metals (ZnO, Hg) that are dangerous to living things and do not biodegrade.^{171–173} Zhao et al.¹⁷⁴ proposed a soft measurement model using the lion-swarm-optimizer-based extreme learning machine (LSO-ELM) to enhance soft quality assessment in wastewater treatment processes with acceptable prediction accuracy, offering more accessible resilience, calibration, and the ability to handle complex and nonlinear

data sets.¹⁷⁵ ANN modeling configurations may additionally be employed to ascertain the performance processes of wastewater treatment plants and forecast BOD. Online control systems and time savings are two benefits of using prediction models for estimating BOD.^{30,176} Numerous ANN algorithms, including ensemble ANN (EANN) and single AAN (SANN), conjugate gradient, feed-forward backpropagation (FFBP) algorithm, gradient descent, Broyden Fletcher Goldfarb Shanno (BFGS), conjugate gradient, radial basis function neural networks (RBFNN), neural network fitting (NNF), cascade forward back-propagation (CFBP), early warning algorithm, vector distance algorithm (VDA), and proportional-integral-derivative (PID) algorithm, have been used to predict and monitor water quality parameters with satisfactory results.¹⁷⁷ One of the most important dangerous components that must be measured in wastewater is copper (Cu) ion. Bhagat et al.¹⁷⁸ have developed three AI models for predicting copper ion adsorption including SVM, ANN, and Grid-RF (random forest based on grid optimization). Numerous methods have been investigated to assess, monitor, and enhance surface water quality because it is necessary for human endeavors such as industry, agriculture, and raising animals. In this direction, Yan et al.¹⁷⁹ examined the use of ML techniques in conjunction with WQIs, trophic status indices (TSIs), and heavy metal indices (HMLIs) to evaluate surface water quality.

Table 3. Real-World Examples of AI- and IoT-Based Water Quality Assessment Projects

Project Name	Sensing parameter	Sensing Technology	Location	ref.
Microsoft Clean Water AI	Harmful bacteria and particles	Neural network model	All over world	188
IBM Watson Water	Dissolved organic carbon (DOC)	IoT, AI, remote sensing, and geospatial analytics	Villages across India	189
Aqueduct Water Risk Atlas	Risk factors - drought, floods, and pollution	AI	West Asia and North Africa	190
Tata Water Mission (TWM) by Tata Group	Water, sanitation, and hygiene (WaSH)	AI and IoT	India	191
Zwemwater app	Swimming	AI	Netherlands	192
Waternet water utility	Drinking water	retrospective analysis (City Blueprint Approach)	Amsterdam	193
Environmental Protection Agency (EPA)	temperature, salinity, and dissolved oxygen levels	sensors and AI	Chesapeake Bay	194
One Water	Weather change, water pollution	IoT and AI	Los Angeles	195
Groundwater	hexavalent chromium	Autoregressive Integrated Moving Average (ARIMA)-based models, Long Short-Term Memory-based framework	Hanford Site in Washington State	196, 197
Drinking water	Radiological and nonradiological	-	Hanford Site in Washington State	198
IBM volunteers and The Nature Conservancy India	Dissolved Oxygen	IoT, IBM Geospatial Analytic	Lake Sembakkam	199
Dissolved oxygen	pH, BOD, water temperature, DO)	ANN, ANFIS	Mathura, India (Yamuna River)	200
Tabriz wastewater treatment plan	Daily influent BOD/COD, TSS, pH, previous BOD/COD effluent	ANN, ANFIS, SVM	Tabriz, Iran	201
Arizona utility company	Blue-green algae conc., conductivity, chlorophyll conc., DO, pH, turbidity	Microbial potentiometric sensor and RNN (LSTM)	Mesa, Arizona (algae cultivation pond)	202

The evaluation of various machine learning approaches for water quality monitoring is crucial. Conventional methods like regression models are computationally efficient but struggle to capture nonlinear correlations in data. Decision tree-based techniques like RF and Gradient Boosting are suitable for classification but need adjustments to avoid overfitting. Support Vector Machines (SVMs) are effective for small data sets but can be computationally expensive for big or high-dimensional data. Neural networks (NN) offer unparalleled accuracy for complex pattern modeling and time-series prediction, but their applicability in resource-constrained environments is limited by large data sets and computer power. Unsupervised methods like clustering algorithms are useful for detecting anomalies but rely heavily on domain knowledge and parameter tweaking. Combining methods like neural networks for pattern recognition and clustering for anomaly identification is the most reliable approach.

Water data and analytics are increasingly relevant in the workplace, with numerous data sources and business intelligence tools available. It is crucial to incorporate these data sources in a comprehensive, organizational-wide monitoring plan for water quality. Implementing a metrics-driven approach for data handling can improve operations and increase savings, thereby enhancing overall efficiency.

6. INTEGRATION OF AI AND IOT FOR COMPREHENSIVE MONITORING

AI and IoT can significantly improve the accuracy, efficacy, and efficiency of monitoring and management activities in water quality evaluation. By integrating these strategies, organizations and stakeholders can harness the full potential of AI and IoT for water quality assessment, leading to improved water management and environmental conservation. Data collection is the first step, followed by data inspection to identify anomalies and outliers. Data preprocessing involves standard data reduction, cleansing, and transformation, while repetitive and irrelevant input reductions help reduce data. AI-powered

sensors can gather data on water quality aspects like temperature, turbidity, pH, and contaminants presence. These sensors can be placed in key areas like lakes, reservoirs, and rivers to provide an overall view of water quality. AI-based data analytics can identify patterns and trends associated with water quality, such as pollution or dangerous algal blooms. This data can be used to create plans for preserving water quality and reducing pollution effects. Reports or dashboards can also be created to compile data on water quality and highlight patterns, enabling public, corporate, and government representatives to make informed decisions about water quality management.^{180,181}

Real-time sensing data gathering and analysis of generated data offers several benefits, including reduced environmental impacts and more sustainable water use practices.^{116,182} Real-time data offers a comprehensive view of water quality, aiding in risk assessment and decision-making. It aids in early stage estimation of water quality issues, detecting pollutants before they reach drinking water supplies, and predicting climate change's impact on water resources.¹⁸³ Placing sensors in strategic locations may offer the advantage of less manual data collecting and processing required. Monitoring the water quality regularly is a crucial practice that may aid in tracking the hydrological data of water bodies.¹⁸⁴ IoT technologies and remote monitoring allow for real-time data access. A Belief Rule Based (BRB) system, DL neural network models, and Spark streaming analysis using Spark MLlib may all be used to visualize data gathered at a separate site on a server PC. The agent will receive an automatic warning SMS alert if the obtained value exceeds the threshold amount.¹²⁵ Some of the real-world examples of AI and IoT-based water quality assessment projects are listed in Table 3. The CANARY event detection program aims to enhance the identification of contaminated water distribution networks by analyzing sensor data in real-time. When used with strategically positioned water quality sensors, it improves detection probability and speed by spotting anomalies and alerting operators when a

possible contaminant is present.^{185,186} This system provides early warning for pollution in urban water sources and helps quickly detect unusual circumstances. The sensitivity of the model depends on optimizing configuration settings, including adjusting BED parameters to increase or decrease alarm generation chances.¹⁸⁷

7. CASE STUDIES AND PRACTICAL IMPLEMENTATIONS

With the help of Landsat 8 OLI (L8 OLI), Sentinel 2A (S2A), and Göktürk-2 (GK2) satellite sensors, a study was carried out by Batur et al.²⁰³ to predict the WQI of Lake Gala (Turkey) for parameters like pH, dissolved oxygen, total suspended solids, total dissolved substance, and dissolved oxygen using satellite image fusion based on principal component analysis (PCA) data fusion and mining techniques. In just four months, the new Remote Monitoring and Management (RMM) platform created by a local tech firm RefillBot has been successfully implemented throughout 100 villages in the Karnataka state of India. The platform uses advanced sensors and IoT technologies that allow real-time 24/7 water quality monitoring.^{191,204} Yan et al.²⁰⁵ recommended using particle swarm optimization (PSO) and genetic algorithms to improve the backpropagation (BP) neural network's ability to estimate a lake's oxygen requirement. It was stated that the forecast findings have improved in accuracy. Environmental influences can obscure DO detection in certain circumstances, indicating the current need for models and techniques that can address the drawbacks of conventional sensing approaches.²⁰⁶

A real-time water quality monitoring system was constructed in the Ganga River (India) considering the ecosystem's fragility and importance. The installation of real-time water quality monitoring system (RTWQMS) stations allowed for the real-time measurement of up to 17 parameters by using a variety of sensors to deliver water quality data.^{207,208} Finally, Post et al.²⁰⁹ reported an extremely intriguing finding that may be utilized to identify chemical contaminants, including nitrates, medications, and microplastics. A UV Raman spectrometer is used in the investigation to identify the most common microplastic polymers, nitrate/nitrite, and a few chosen medicines. The findings demonstrate that nitrates and nitrites may be identified and measured; however, because of their uneven distribution, measurements of microplastic particles are hampered. For testing the presence of microplastics in ambient water, a new electrochemical sensor made of chitosan and magnesium oxide nanosheets was presented by Noumani et al.²¹⁰ The sensor exhibits a low LOD for HMT, great sensitivity, and excellent linearity.

The primary reason why water monitoring for oil and gas presents special difficulties is that the water is never uniform. KETOS can help oil and gas companies keep track of their water quality from process to process and well to well. KETOS can monitor in real-time allowing for the flexibility to test regularly and often so that any spikes or dramatic changes in water quality can be caught early and dealt with immediately in a way that was not feasible for operations relying on third-party sampling and testing. The KETOS SHIELD removes the need for human intervention.²¹¹ KETOS SHIELD is a smart water management system that uses a single IoT (Internet-aware) modular device to give operators lab-accurate findings on dozens of parameters, such as nutrients, heavy metals, and inorganics.²¹¹

8. ADVANCEMENTS IN AI AND IOT TECHNOLOGIES

Recently, there have been a few developments in water quality assessment based on AI algorithms and ML. Researchers utilized pre-trained neural networks for tasks such as water quality prediction, speeding up model training, and improving performance, especially in data-scarce regions. Hybrid models took advantage of both domain knowledge and large data sets to improve the water quality predictions.²¹² Ajkawe et al.²¹³ connected intelligence (CI), connected innovation intelligence (CII), and the AI learning paradigm were used to solve the basic flaws in the current legacy IoT-based WQMS designs to provide value and strengthen the dependability of the system's decision-making capabilities. The water quality evaluation relies on using ensemble learning techniques that integrate several ML models to increase precision. Local data processing using fog computing and IoT devices reduced latency and the requirement for lengthy data transfer.^{214,215}

Water quality metrics can be estimated using modeling and data mining, with advancements in data analytics and machine learning playing a significant role. Algorithmic responses can be an inexpensive substitute for hardware sensors, enhancing the accuracy and accessibility of water quality assessment. The use of IoT devices and sensors is transforming water quality assessment and management, making systems more accessible and accurate in various settings. The increasing popularity of low-cost, reliable sensors is making them more common. New IoT sensors can measure multiple aspects of water quality simultaneously, and research is being conducted on blockchain technologies to ensure data confidentiality and integrity. Lin et al.²¹⁶ utilized GIS and blockchain technologies to identify contaminated units at irrigation intakes and track pollution. The encrypted data could be beneficial for real-world water quality management. The proposed blockchain traceability system and GIS tools could focus on pollution sources and irrigation unit paths.^{32,217–219} IoT devices can send real-time alarms and notifications when water quality metrics fall outside safe ranges. Energy-efficient sensor designs are emerging, enabling extended battery life and reduced environmental impact, especially in remote monitoring scenarios.

Smart Water Quality Monitoring (SWQM) is a method used to measure water quality parameters using servers, routing devices, IoT sensors, and other hardware in an IoT ecosystem. It supports tasks like system activation, security communication, action specifications, and detection. The top five hardware parts of SWQM include pH, ultrasonic, digital thermometer, and RF module for turbidity sensors. SWQM is useful for in-home applications, municipal waste recycling, agriculture, and aquaculture, as well as monitoring water quality in rivers and lakes.²²⁰ IoT water flow sensors are useful for quickly identifying leaks since they measure the rate at which water flows through a pipe. A variation in the standard rate detected by sensor data may be an indication of a leak in the pipe or any other operational issue. Part et al.²²¹ created an ensemble ML model (XGBoost) to forecast how quickly water quality would recover in a water treatment facility following a disruption. Pretreatment increased the XGBoost model's root-mean-square error—observation standard deviation ratio from 0.730 to 0.373. An emerging field of ML modeling called explainable artificial intelligence (XAI) offers a logical explanation of model performance. A further study²²² utilized 18 input variables to create an XGBoost ensemble ML model that predicted the concentration of chlorophyll-a (Chl-a) in

water. Apart from using XGBoost, Rodríguez-López et al.²²³ found that LightGBM and AdaBoost models showed better performance for forecasting inland aquatic habitats' water quality. Wu et al.²²⁴ developed a multitask DL-based water quality prediction model with the COD of the Lanzhou part of the Yellow River as the study target. Better information mining from local to full-time series features of water quality is possible using the hybrid CNN-LSTM model. An IoT platform called ThingSpeak is used to monitor and evaluate water quality. It offers analytical tools and visualization using MATLAB programming. Water samples are tested using sensor fusion techniques like TDS and turbidity, and the data is uploaded online to the ThingSpeak platform.²²⁵ Mazumdar et al.²²⁶ used the XGBoost ML model to study groundwater contamination transport, focusing on the common pollutant Cr (VI). They used an enhanced version of the model, which used the Empirical Bayes Search Cross-Validation method for improved prediction. However, due to the complexity of the phenomenon, variability of natural characteristics, and poorly understood interacting mechanisms, many pollutants, including hexavalent chromium, remain challenging to model.

Aquaculture relies on water quality, and real-time monitoring through sensors allows for water parameter prediction. The integration of IoT for AI in aquaculture promotes growth and environmental sustainability. Researchers have effectively used IoT technologies, including big data analysis and AI, to manage and monitor maritime environment quality in the first decade of the 21st century.^{227–230} The mismatch between data quality, gathering, and interpretation in water bodies is due to novel pollutants, spatiotemporal fluctuations, and a lack of standardized methodologies. To address this, a network of multiparametric sensor systems using autonomous vehicles can be deployed, increasing data coverage in space and time. Intelligent algorithms can also be used for forecasting and standardized data analysis.²³¹ The advancement of skilled soft computing, artificial intelligence, and cloud IoT could significantly enhance drinking water quality management systems and reduce the need for operators.²³²

The future of water quality assessment is influenced by technological advancements, environmental concerns, and the growing demand for clean, safe water. The development of advanced sensors, including IoT devices, will enable more precise monitoring of water quality. These sensors will be smaller, durable, and measure a wider range of parameters. The processing of this data will require a combination of AI and big data analytics. Assessing water quality is crucial for protecting ecosystems and biodiversity, reducing the risk of waterborne illnesses, and improving overall water safety.

9. CHALLENGES AND POTENTIAL APPLICATIONS

AI and IoT have the potential to improve water quality evaluation, but they face challenges such as sensor variability, data storage, transmission, and processing. Integrating data from various sources can be complex due to differences in formats, resolutions, and update frequencies. Privacy concerns arise when handling sensitive data, especially in areas with human populations. Ensuring secure data storage, transmission, and access control is crucial. Developing AI models for water quality prediction that generalize well across different locations and environmental conditions is also a challenge.²³³ Challenges such as computational resource limitations, high-speed Internet, and reliable power sources in certain regions

make implementing AI and IoT solutions difficult. Costly deployment of IoT sensors and AI implementation can also pose challenges. AI in water treatment faces issues like data quality, availability, compatibility, interpretability, and multidisciplinary cooperation.²³⁴ The integration of AI and IoT for comprehensive water quality monitoring faces challenges due to variations in data quality and compatibility from different sensors. Scalability is needed to meet the needs of different water utilities and environments, but computationally expensive AI models and significant data storage and processing requirements are also significant. Cyberattacks pose a major concern, as disruptions to water service can have serious consequences for public health and safety. Potential solutions include addressing these issues and ensuring data security.¹⁷²

Industrial applications face challenges such as lack of centralized data on water quality, device dependability, autonomous heavy metal testing, excessive chemical treatment costs, and costly business models. Automation in monitoring and reporting processes through IoT and AI helps water utilities and regulatory agencies ensure water quality standards and regulations more efficiently. Adhering to water quality regulations and standards can be complex, and ensuring AI and IoT systems meet these requirements can be challenging. ML often requires high-quality data in large quantities, which is difficult to obtain due to financial or technological constraints. Existing methods are limited to systems due to complex circumstances in actual water treatment and management systems, restricting the broad applicability of ML techniques. To address these challenges, future engineering and research processes should consider more sophisticated sensors, improve the versatility and dependability of algorithms, and teach multidisciplinary talent with expertise across domains to create more sophisticated ML approaches and implement them in engineering procedures.

The use of IoT and AI in water quality assessment is crucial for preventing contaminated water consumption and reducing waterborne diseases. This is essential for ecosystem health and biodiversity preservation. Research can help understand human activities' impact on water quality and develop strategies to mitigate these effects. Clean drinking water reduces waterborne illness risk, and AI and IoT can improve water resource management, reduce waste, and optimize utilization. Real-time data can assist emergency responders in making informed decisions to protect public safety. Future aquaculture operations will benefit from new technology, reducing human labor and promoting efficient maintenance and resource use.

10. CONCLUSIONS

AI and IoT-supported 6GS technologies represent a significant advancement in water quality assessment, providing real-time data on water quality parameters, early warnings, and predictive analytics.²³⁵ AI integrates data from multiple sources, improving the accuracy of assessments and management decisions. ML algorithms for anomaly detection trigger alerts, allowing swift responses to potential threats. IoT technologies enable remote monitoring and control of water quality parameters, making data more accessible and actionable for decision-makers. The integration of cutting-edge technologies with bionanotechnology-based water purification solutions can transform current methods. Platform urbanism is emerging as a key instrument for smart urban governance and creating a new version of cities, known as City 4.0.²³⁶

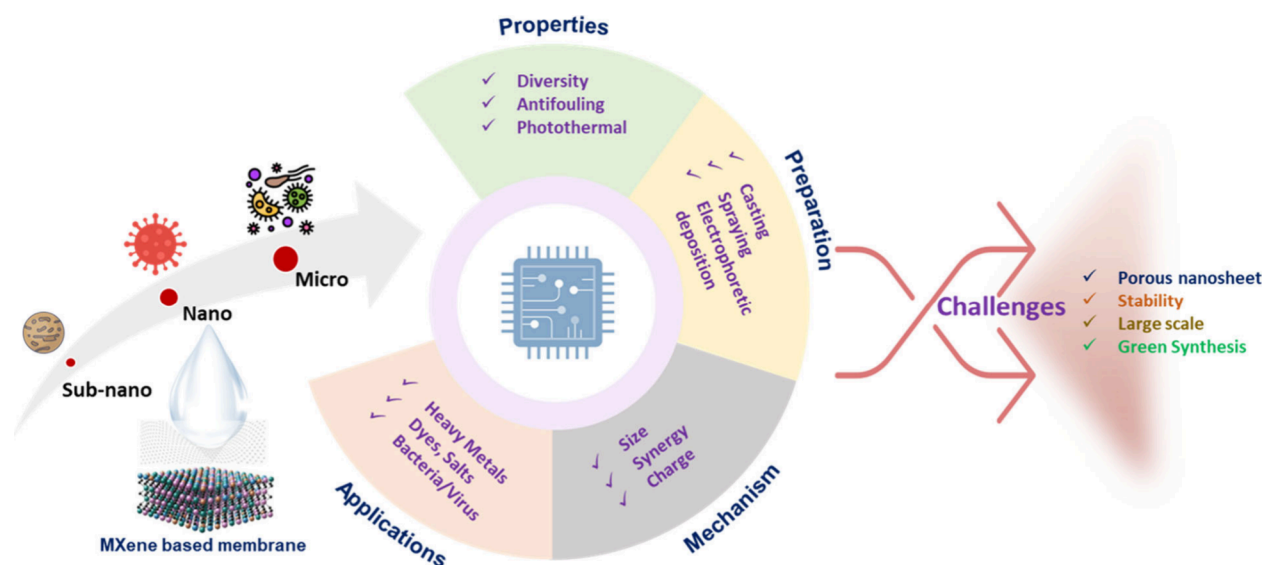


Figure 8. A futuristic concept of AI-enabled MXene-based membrane technology for water management.

AI-enabled systems can significantly enhance the effectiveness of 2D layered transition metal dichalcogenides (TMDCs) such as MXene and tungsten disulfide (WS_2) based membrane technology for water management.^{237–239} MXenes, a class of two-dimensional materials, are known for their excellent properties and application in water purification and desalination. Integrating AI into the system can improve the design, operation, and monitoring of MXene-based membrane processes for efficient water treatment. This system can become more efficient, cost-effective, and adaptable to various water treatment challenges. This approach can help address global water scarcity issues, reduce energy consumption in water treatment, and improve water quality. In this regard, an AI-driven MXene-based membrane system is proposed for water quality measurement (shown in Figure 8). Monitoring the health of aquatic ecosystems and studying the effects of industrial processes, agriculture, and urban development on water quality are critical applications. Timely and accurate water quality assessment using AI and IoT technologies is essential for protecting public health.

AUTHOR INFORMATION

Corresponding Authors

Kamil Reza Khondakar – School of Science, Woxsen University, Hyderabad, Telangana 502345, India; orcid.org/0000-0001-5136-622X; Email: kamilreza@gmail.com, kamil.reza@woxsen.edu.in

Hirak Mazumdar – Department of Computer Science and Engineering, Adamas University, Kolkata, West Bengal 700126, India; orcid.org/0000-0001-6469-1515; Email: hirak.mazumdar1@adamasuniversity.ac.in, hirakm.tech.ece@gmail.com

Ajeet Kaushik – Nano BioNTechs Laboratory, Department of Environmental Engineering, Florida Polytechnic University, Lakeland, Florida 33805, United States; orcid.org/0000-0003-4206-1541; Email: akaushik@floridapoly.edu, ajeet.npl@gmail.com

Authors

Suparna Das – Department of Computer Science and Engineering, BVRIT HYDERABAD College of Engineering for Women, Hyderabad, Telangana 500090, India

Yogendra Kumar Mishra – Smart Materials, NanoSYD, Mads Clausen Institute, University of Southern Denmark, 6400 Sønderborg, Denmark; orcid.org/0000-0002-8786-9379

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acsestwater.4c00360>

Author Contributions

†CRediT: **Suparna Das** conceptualization, formal analysis, methodology, writing - original draft; **Kamil reza Khondakar** conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, supervision, validation, visualization, writing - original draft, writing - review & editing; **Hirak Mazumdar** conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, supervision, validation, visualization, writing - original draft, writing - review & editing; **Ajeet Kumar Kaushik** conceptualization, investigation, methodology, project administration, resources, supervision, validation, writing - review & editing; **Yogendra Kumar Mishra** investigation, project administration, resources, supervision, writing - review & editing.

Author Contributions

SD: Writing—original draft. HM: Writing—original draft and Visualization. KK: Writing—review and editing. YKM: Writing—review and editing. AK: Writing—review and editing. CRediT: **Suparna Das** conceptualization, formal analysis, methodology, writing - original draft; **Kamil reza Khondakar** conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, supervision, validation, visualization, writing - original draft, writing - review & editing; **Hirak Mazumdar** conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, supervision, validation, visualization, writing - original draft, writing - review & editing; **Ajeet Kumar Kaushik** conceptualization, investigation, methodology, project administration, resources, supervision, validation, writing - review & editing.

tion, writing - review & editing; **Yogendra Kumar Mishra** investigation, project administration, resources, supervision, writing - review & editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

K.M. acknowledges the funding from the ESS lighthouse on hard materials in 3D, SOLID (Danish Agency for Science and Higher Education, grant number 8144-00002B), Denmark, and NANOChem (national infrastructure UFM 52290-00010B, NANOChem, Denmark), from BHJ Fonden and Fabrikant Mads Clausen Fonden, Denmark.

REFERENCES

- (1) Drinking-water. <https://www.who.int/news-room/fact-sheets/detail/drinking-water> (accessed 2023-09-09).
- (2) Fathalla, M.; Selvaganapathy, P. R. Colorimetric Detection of Heavy Metal Ions Using Superabsorptive Hydrogels and Evaporative Concentration for Water Quality Monitoring. *ACS EST Water* **2022**, *2* (4), 658–666.
- (3) Korostynska, O.; Mason, A.; Al-Shamma'a, A. I. Monitoring Pollutants in Wastewater: Traditional Lab Based versus Modern Real-Time Approaches. In *Smart Sensors for Real-Time Water Quality Monitoring*; Mukhopadhyay, S. C., Mason, A., Eds.; Smart Sensors, Measurement and Instrumentation; Springer: Berlin, Heidelberg, 2013; pp 1–24. DOI: 10.1007/978-3-642-37006-9_1.
- (4) Shi, Z.; Chow, C. W. K.; Fabris, R.; Liu, J.; Jin, B. Applications of Online UV-Vis Spectrophotometer for Drinking Water Quality Monitoring and Process Control: A Review. *Sensors* **2022**, *22* (8), 2987.
- (5) Xu, K.; Chen, Q.; Zhao, Y.; Ge, C.; Lin, S.; Liao, J. Cost-Effective, Wireless, and Portable Smartphone-Based Electrochemical System for on-Site Monitoring and Spatial Mapping of the Nitrite Contamination in Water. *Sens. Actuators, B* **2020**, *319*, No. 128221.
- (6) Thio, S. K.; Lee, S.; Bae, S.; Park, S.-Y. Lab on a Smartphone (LOS): A Low-Cost Portable Platform for Real-Time on-Site Water Quality Detection. *2018 International Conference on Optical MEMS and Nanophotonics (OMN)*; 2018, pp 1–2.
- (7) Zhao, C.; Chen, L.; Zhong, G.; Wu, Q.; Liu, J.; Liu, X. A Portable Analytical System for Rapid On-Site Determination of Total Nitrogen in Water. *Water Res.* **2021**, *202*, 117410.
- (8) Cho, J. H.; Gao, Y.; Choi, S. A Portable, Single-Use, Paper-Based Microbial Fuel Cell Sensor for Rapid, On-Site Water Quality Monitoring. *Sensors* **2019**, *19* (24), 5452.
- (9) Lin, J.-Y.; Tsai, H.-L.; Lyu, W.-H. An Integrated Wireless Multi-Sensor System for Monitoring the Water Quality of Aquaculture. *Sensors (Basel)* **2021**, *21* (24), 8179.
- (10) Wong, B. P.; Kerkez, B. Real-Time Environmental Sensor Data: An Application to Water Quality Using Web Services. *Environmental Modelling & Software* **2016**, *84*, 505–517.
- (11) Chung, W.-Y.; Yoo, J.-H. Remote Water Quality Monitoring in Wide Area. *Sens. Actuators, B* **2015**, *217*, 51–57.
- (12) Wang, J.; Ren, X.; Shen, Y.; Liu, S. A Remote Wireless Sensor Networks for Water Quality Monitoring. *2010 International Conference on Innovative Computing and Communication and 2010 Asia-Pacific Conference on Information Technology and Ocean Engineering* **2010**, 7–12.
- (13) Das, B.; Jain, P. C. Real-Time Water Quality Monitoring System Using Internet of Things. *2017 International Conference on Computer, Communications and Electronics (Comptelix)* **2017**, 78–82.
- (14) Cloete, N. A.; Malekian, R.; Nair, L. Design of Smart Sensors for Real-Time Water Quality Monitoring. *IEEE Access* **2016**, *4*, 3975–3990.
- (15) Wang, Z.; Wang, Q.; Hao, X. The Design of the Remote Water Quality Monitoring System Based on WSN. *2009 5th International Conference on Wireless Communications, Networking and Mobile Computing* **2009**, 1–4.
- (16) Andres, L.; Boateng, K.; Borja-Vega, C.; Thomas, E. A Review of In-Situ and Remote Sensing Technologies to Monitor Water and Sanitation Interventions. *Water* **2018**, *10* (6), 756.
- (17) Harwin, S.; Lucieer, A. Assessing the Accuracy of Georeferenced Point Clouds Produced via Multi-View Stereopsis from Unmanned Aerial Vehicle (UAV) Imagery. *Remote Sensing* **2012**, *4* (6), 1573–1599.
- (18) Jaywant, S. A.; Arif, K. M. A Comprehensive Review of Microfluidic Water Quality Monitoring Sensors. *Sensors* **2019**, *19* (21), 4781.
- (19) Saez, J.; Catalan-Carrio, R.; Owens, R. M.; Basabe-Desmonts, L.; Benito-Lopez, F. Microfluidics and Materials for Smart Water Monitoring: A Review. *Anal. Chim. Acta* **2021**, *1186*, No. 338392.
- (20) Jang, A.; Zou, Z.; Lee, K. K.; Ahn, C. H.; Bishop, P. L. State-of-the-Art Lab Chip Sensors for Environmental Water Monitoring. *Meas. Sci. Technol.* **2011**, *22* (3), No. 032001.
- (21) Li, Z.; Liu, H.; Wang, D.; Zhang, M.; Yang, Y.; Ren, T. Recent Advances in Microfluidic Sensors for Nutrients Detection in Water. *TrAC Trends in Analytical Chemistry* **2023**, *158*, No. 116790.
- (22) Das, S.; Mazumdar, H. Integration of Artificial Intelligence and Computational Thinking in Lab-on-a-Chip Technology for Quality Improvement in Healthcare *Lab-on-a-Chip Devices for Advanced Biomedicine*; Royal Society of Chemistry, 2024.
- (23) Dey, S.; Vaidyanathan, R.; Reza, K. K.; Wang, J.; Wang, Y.; Nel, H. J.; Law, S.-C.; Tyler, J.; Rossjohn, J.; Reid, H. H.; Ibn Sina, A. A.; Thomas, R.; Trau, M. A Microfluidic-SERS platform for Isolation and Immuno-Phenotyping of Antigen Specific T-Cells. *Sens. Actuators, B* **2019**, *284*, 281–288.
- (24) Khondakar, K. R.; Kaushik, A. K. *Nanotechnology in Cancer Management: Precise Diagnostics toward Personalized Health Care*; Elsevier, 2021.
- (25) Kumar, P.; Shimali; Chamoli, S.; Khondakar, K. R. Advances in Optical and Electrochemical Sensing of Bisphenol a (BPA) Utilizing Microfluidic Technology: A Mini Perspective. *Methods* **2023**, *220*, 69–78.
- (26) Mazumdar, H.; Khondakar, K. R.; Das, S.; Halder, A.; Kaushik, A. Artificial Intelligence for Personalized Nanomedicine; from Material Selection to Patient Outcomes. *Expert Opinion on Drug Delivery* **2025**, *22* (1), 85–108.
- (27) Das, S.; Mazumdar, H.; Khondakar, K. R.; Kaushik, A. Machine Learning Integrated Graphene Oxide-Based Diagnostics, Drug Delivery, Analytical Approaches to Empower Cancer Diagnosis. *BMEMat* **2024**, No. e12117.
- (28) Digital Arsenic Test Kit - Palintest. <http://www.palintest.com/products/arsenator/> (accessed 2024-04-19).
- (29) Arora, N. K.; Mishra, I. Sustainable Development Goal 6: Global Water Security. *Environmental Sustainability* **2022**, *5* (3), 271–275.
- (30) Ighalo, J. O.; Adeniyi, A. G.; Marques, G. Artificial Intelligence for Surface Water Quality Monitoring and Assessment: A Systematic Literature Analysis. *Model. Earth Syst. Environ.* **2021**, *7* (2), 669–681.
- (31) Mehmood, H.; Liao, D.; Mahadeo, K. A Review of Artificial Intelligence Applications to Achieve Water-Related Sustainable Development Goals. *2020 IEEE/ITU International Conference on Artificial Intelligence for Good (AI4G)* **2020**, 135–141.
- (32) Dogo, E. M.; Nwulu, N. I.; Twala, B.; Aigbavboa, C. A Survey of Machine Learning Methods Applied to Anomaly Detection on Drinking-Water Quality Data. *Urban Water Journal* **2019**, *16* (3), 235–248.
- (33) Singh, A.; Dhau, J.; Kumar, R.; Badru, R.; Singh, P.; Kumar Mishra, Y.; Kaushik, A. Tailored Carbon Materials (TCM) for Enhancing Photocatalytic Degradation of Polyaromatic Hydrocarbons. *Prog. Mater. Sci.* **2024**, *144*, No. 101289.

- (34) Russo, T.; Alfredo, K.; Fisher, J. Sustainable Water Management in Urban, Agricultural, and Natural Systems. *Water* **2014**, *6* (12), 3934–3956.
- (35) McDonald, R. I.; Weber, K.; Padowski, J.; Flörke, M.; Schneider, C.; Green, P. A.; Gleeson, T.; Eckman, S.; Lehner, B.; Balk, D.; Boucher, T.; Grill, G.; Montgomery, M. Water on an Urban Planet: Urbanization and the Reach of Urban Water Infrastructure. *Global Environmental Change* **2014**, *27*, 96–105.
- (36) Bogardi, J. J.; Leentvaar, J.; Sebesvári, Z. *Biologia Futura: Integrating Freshwater Ecosystem Health in Water Resources Management*. *BIOLOGIA FUTURA* **2020**, *71* (4), 337–358.
- (37) Stefanakis, A. I. The Role of Constructed Wetlands as Green Infrastructure for Sustainable Urban Water Management. *Sustainability* **2019**, *11* (24), 6981.
- (38) Glavan, M. *Water Challenges of an Urbanizing World*; BoD – Books on Demand, 2018.
- (39) Aljamali, N. M.; Jawad, A. M.; Alabbasy, A. J.; Salih, A. M. A Literature Review on Types of Contamination (Biological, Chemical, Medical). *Int. J. Green Chem.* **2019**, *5*, 7–14.
- (40) Kordbacheh, F.; Heidari, G. Water Pollutants and Approaches for Their Removal. *Materials Chemistry Horizons* **2023**, *2* (2), 139–153.
- (41) Liu, G.; Zhang, Y.; Knibbe, W.-J.; Feng, C.; Liu, W.; Medema, G.; van der Meer, W. Potential Impacts of Changing Supply-Water Quality on Drinking Water Distribution: A Review. *Water Res.* **2017**, *116*, 135–148.
- (42) Madhav, S.; Ahamad, A.; Singh, A. K.; Kushawaha, J.; Chauhan, J. S.; Sharma, S.; Singh, P. Water Pollutants: Sources and Impact on the Environment and Human Health. In *Sensors in Water Pollutants Monitoring: Role of Material*; Pooja, D., Kumar, P., Singh, P., Patil, S., Eds.; Advanced Functional Materials and Sensors; Springer: Singapore, 2020; pp 43–62.
- (43) Okoye, C. O.; Nyaruaba, R.; Ita, R. E.; Okon, S. U.; Addey, C. I.; Ebido, C. C.; Opabunmi, A. O.; Okeke, E. S.; Chukwudozie, K. I. Antibiotic Resistance in the Aquatic Environment: Analytical Techniques and Interactive Impact of Emerging Contaminants. *Environmental Toxicology and Pharmacology* **2022**, *96*, No. 103995.
- (44) Palansooriya, K. N.; Yang, Y.; Tsang, Y. F.; Sarkar, B.; Hou, D.; Cao, X.; Meers, E.; Rinklebe, J.; Kim, K.-H.; Ok, Y. S. Occurrence of Contaminants in Drinking Water Sources and the Potential of Biochar for Water Quality Improvement: A Review. *Critical Reviews in Environmental Science and Technology* **2020**, *50* (6), 549–611.
- (45) *Sensors in Water Pollutants Monitoring: Role of Material*; Pooja, D., Kumar, P., Singh, P., Patil, S., Eds.; Advanced Functional Materials and Sensors; Springer: Singapore, 2020.
- (46) Rodriguez-Narvaez, O. M.; Peralta-Hernandez, J. M.; Goonetilleke, A.; Bandala, E. R. Treatment Technologies for Emerging Contaminants in Water: A Review. *Chemical Engineering Journal* **2017**, *323*, 361–380.
- (47) Summers, J. K. *Water Quality: Science, Assessments and Policy*; BoD – Books on Demand, 2020.
- (48) Gogoi, A.; Mazumder, P.; Tyagi, V. K.; Tushara Chaminda, G. G.; An, A. K.; Kumar, M. Occurrence and Fate of Emerging Contaminants in Water Environment: A Review. *Groundwater for Sustainable Development* **2018**, *6*, 169–180.
- (49) Gonsioroski, A.; Mourikes, V. E.; Flaws, J. A. Endocrine Disruptors in Water and Their Effects on the Reproductive System. *International Journal of Molecular Sciences* **2020**, *21* (6), 1929.
- (50) Sonone, S.; Jadhav, S.; Singh Sankhla, M.; Kumar, R. Water Contamination by Heavy Metals and Their Toxic Effect on Aquaculture and Human Health through Food Chain. *Letters in Applied NanoBioScience* **2020**, *10*, 2148–2166.
- (51) Tiwari, A. K.; Pal, D. B. Chapter 11 - Nutrients Contamination and Eutrophication in the River Ecosystem. In *Ecological Significance of River Ecosystems*; Madhav, S., Kanhaiya, S., Srivastav, A., Singh, V., Singh, P., Eds.; Elsevier, 2022; pp 203–216.
- (52) Bottoni, P.; Caroli, S.; Caracciolo, A. B. Pharmaceuticals as Priority Water Contaminants. *Toxicol. Environ. Chem.* **2010**, *92* (3), 549–565.
- (53) Chaturvedi, P.; Shukla, P.; Giri, B. S.; Chowdhary, P.; Chandra, R.; Gupta, P.; Pandey, A. Prevalence and Hazardous Impact of Pharmaceutical and Personal Care Products and Antibiotics in Environment: A Review on Emerging Contaminants. *Environmental Research* **2021**, *194*, No. 110664.
- (54) Babič, M. N.; Gunde-Cimerman, N.; Vargha, M.; Tischner, Z.; Magyar, D.; Verissimo, C.; Sabino, R.; Viegas, C.; Meyer, W.; Brandão, J. Fungal Contaminants in Drinking Water Regulation? A Tale of Ecology, Exposure, Purification and Clinical Relevance. *International Journal of Environmental Research and Public Health* **2017**, *14* (6), 636.
- (55) Björklund, G.; Semenova, Y.; Pivina, L.; Dadar, M.; Rahman, Md. M.; Aaseth, J.; Chirumbolo, S. Uranium in Drinking Water: A Public Health Threat. *Arch. Toxicol.* **2020**, *94* (5), 1551–1560.
- (56) Li, P.; Karunanidhi, D.; Subramani, T.; Srinivasamoorthy, K. Sources and Consequences of Groundwater Contamination. *Arch. Environ. Contam. Toxicol.* **2021**, *80* (1), 1–10.
- (57) Lamine, S.; Petropoulos, G. P.; Brewer, P. A.; Bachari, N.-E.-I.; Srivastava, P. K.; Manevski, K.; Kalaitzidis, C.; Macklin, M. G. Heavy Metal Soil Contamination Detection Using Combined Geochemistry and Field Spectroradiometry in the United Kingdom. *Sensors (Basel)* **2019**, *19* (4), 762.
- (58) Yusuf, A.; Sadiq, A.; Giwa, A.; Eke, J.; Pikuda, O.; Eniola, J. O.; Ajiwokewu, B.; Sambudi, N. S.; Bilad, M. R. Updated Review on Microplastics in Water, Their Occurrence, Detection, Measurement, Environmental Pollution, and the Need for Regulatory Standards. *Environ. Pollut.* **2022**, *292*, No. 118421.
- (59) Kroehler, C. J. Potable Water Quality Standards and Regulations: A Historical and World Overview. In *Potable Water: Emerging Global Problems and Solutions*; Yunus, T., Grady, C. A., Eds.; Springer International Publishing: Cham, 2014; pp 1–36.
- (60) Begum, W.; Goswami, L.; Sharma, B. B.; Kushwaha, A. Assessment of Urban River Pollution Using the Water Quality Index and Macro-Invertebrate Community Index. *Environ. Dev. Sustain* **2023**, *25* (8), 8877–8902.
- (61) Valenzuela, E. F.; Menezes, H. C.; Cardeal, Z. L. Passive and Grab Sampling Methods to Assess Pesticide Residues in Water. A Review. *Environ. Chem. Lett.* **2020**, *18* (4), 1019–1048.
- (62) Alam, A. U.; Clyne, D.; Jin, H.; Hu, N.-X.; Deen, M. J. Fully Integrated, Simple, and Low-Cost Electrochemical Sensor Array for in Situ Water Quality Monitoring. *ACS Sens.* **2020**, *5* (2), 412–422.
- (63) Hoang, H.-G.; Lin, C.; Tran, H.-T.; Chiang, C.-F.; Bui, X.-T.; Cheruiyot, N. K.; Shern, C.-C.; Lee, C.-W. Heavy Metal Contamination Trends in Surface Water and Sediments of a River in a Highly-Industrialized Region. *Environmental Technology & Innovation* **2020**, *20*, No. 101043.
- (64) Amrita, C.; Babiyola, D. Analysing the Water Quality Parameters from Traditional to Modern Methods in Aquaculture. *Int. J. Sci., Environ.* **2018**, *7*, 1954–1961.
- (65) Yang, L.; Driscoll, J.; Sarigai, S.; Wu, Q.; Lippitt, C. D.; Morgan, M. Towards Synoptic Water Monitoring Systems: A Review of AI Methods for Automating Water Body Detection and Water Quality Monitoring Using Remote Sensing. *Sensors* **2022**, *22* (6), 2416.
- (66) Kamil Reza, K.; Singh, M. K.; Yadav, S. K.; Singh, J.; Agrawal, V. V.; Malhotra, B. D. Quantum Dots Based Platform for Application to Fish Freshness Biosensor. *Sens. Actuators, B* **2013**, *177*, 627–633.
- (67) Khondakar, K. R.; Kaushik, A. Role of Wearable Sensing Technology to Manage Long COVID. *Biosensors* **2023**, *13* (1), 62.
- (68) Nehra, M.; Dilbaghi, N.; Marrazza, G.; Kaushik, A.; Sonne, C.; Kim, K.-H.; Kumar, S. Emerging Nanobiotechnology in Agriculture for the Management of Pesticide Residues. *Journal of Hazardous Materials* **2021**, *401*, No. 123369.
- (69) Reza, K. K.; Ali, Md. A.; Srivastava, S.; Agrawal, V. V.; Biradar, A. M. Tyrosinase Conjugated Reduced Graphene Oxide Based Biointerface for Bisphenol A Sensor. *Biosens. Bioelectron.* **2015**, *74*, 644–651.
- (70) Chee, G.-J.; Nomura, Y.; Ikebukuro, K.; Karube, I. Optical Fiber Biosensor for the Determination of Low Biochemical Oxygen Demand. *Biosens. Bioelectron.* **2000**, *15*, 371–376.

- (71) Hossain, S. M. Z.; Mansour, N. Biosensors for On-Line Water Quality Monitoring – a Review. *Arab Journal of Basic and Applied Sciences* **2019**, *26* (1), 502–518.
- (72) Hara, T. O.; Singh, B. Electrochemical Biosensors for Detection of Pesticides and Heavy Metal Toxicants in Water: Recent Trends and Progress. *ACS EST Water* **2021**, *1* (3), 462–478.
- (73) Lukyanenko, K. A.; Denisov, I. A.; Sorokin, V. V.; Yakimov, A. S.; Esimbekova, E. N.; Belobrov, P. I. Handheld Enzymatic Luminescent Biosensor for Rapid Detection of Heavy Metals in Water Samples. *Chemosensors* **2019**, *7* (1), 16.
- (74) Rainbow, J.; Sedlackova, E.; Jiang, S.; Maxted, G.; Moschou, D.; Richtera, L.; Estrela, P. Integrated Electrochemical Biosensors for Detection of Waterborne Pathogens in Low-Resource Settings. *Biosensors* **2020**, *10* (4), 36.
- (75) Guo, Y.; Wang, J.; Shinde, S.; Wang, X.; Li, Y.; Dai, Y.; Ren, J.; Zhang, P.; Liu, X. Simultaneous Wastewater Treatment and Energy Harvesting in Microbial Fuel Cells: An Update on the Biocatalysts. *RSC Adv.* **2020**, *10*, 25874–25887.
- (76) Roy, H.; Rahman, T. U.; Tasnim, N.; Arju, J.; Rafid, Md. M.; Islam, Md. R.; Pervaz, Md. N.; Cai, Y.; Naddeo, V.; Islam, Md. S. Microbial Fuel Cell Construction Features and Application for Sustainable Wastewater Treatment. *Membranes (Basel)* **2023**, *13* (5), 490.
- (77) Li, W.-W.; Yu, H.-Q.; He, Z. Towards Sustainable Wastewater Treatment by Using Microbial Fuel Cells-Centered Technologies. *Energy Environ. Sci.* **2014**, *7* (3), 911–924.
- (78) Cui, Y.; Lai, B.; Tang, X. Microbial Fuel Cell-Based Biosensors. *Biosensors* **2019**, *9* (3), 92.
- (79) Kumar, T.; Naik, S.; Jujjavarappu, S. E. A Critical Review on Early-Warning Electrochemical System on Microbial Fuel Cell-Based Biosensor for on-Site Water Quality Monitoring. *Chemosphere* **2022**, *291*, No. 133098.
- (80) Pasternak, G.; Greenman, J.; Ieropoulos, I. Self-Powered, Autonomous Biological Oxygen Demand Biosensor for Online Water Quality Monitoring. *Sens. Actuators, B* **2017**, *244*, 815–822.
- (81) Chouler, J.; Monti, M. D.; Morgan, W. J.; Cameron, P. J.; Di Lorenzo, M. A Photosynthetic Toxicity Biosensor for Water. *Electrochim. Acta* **2019**, *309*, 392–401.
- (82) Cunha, I.; Biltz, R.; Sales, M. G. F.; Vasconcelos, V. Aptamer-Based Biosensors to Detect Aquatic Phycotoxins and Cyanotoxins. *Sensors* **2018**, *18* (7), 2367.
- (83) Das, S.; Mazumdar, H.; Khondakar, K. R.; Mishra, Y. K.; Kaushik, A. Review—Quantum Biosensors: Principles and Applications in Medical Diagnostics. *ECS Sens. Plus* **2024**, *3* (2), No. 025001.
- (84) McConnell, E. M.; Nguyen, J.; Li, Y. Aptamer-Based Biosensors for Environmental Monitoring. *Frontiers in Chemistry* **2020**, *8*, DOI: 10.3389/fchem.2020.00434.
- (85) Jaywant, S. A.; Arif, K. M. A Comprehensive Review of Microfluidic Water Quality Monitoring Sensors. *Sensors* **2019**, *19* (21), 4781.
- (86) Mohammadi, S.; Nadaraja, A. V.; Roberts, D. J.; Zarifi, M. H. Real-Time and Hazard-Free Water Quality Monitoring Based on Microwave Planar Resonator Sensor. *Sensors and Actuators A: Physical* **2020**, *303*, No. 111663.
- (87) Bickman, S. R.; Campbell, K.; Elliott, C.; Murphy, C.; O’Kennedy, R.; Papst, P.; Lochhead, M. J. An Innovative Portable Biosensor System for the Rapid Detection of Freshwater Cyanobacterial Algal Bloom Toxins. *Environ. Sci. Technol.* **2018**, *52* (20), 11691–11698.
- (88) Chakraborty, U.; Kaur, G.; Rubahn, H.-G.; Kaushik, A.; Chaudhary, G. R.; Mishra, Y. K. Advanced Metal Oxides Nanostructures to Recognize and Eradicate Water Pollutants. *Prog. Mater. Sci.* **2023**, *139*, No. 101169.
- (89) Letsoalo, M. R.; Sithole, T.; Mufamadi, S.; Mazhandu, Z.; Sillanpaa, M.; Kaushik, A.; Mashifana, T. Efficient Detection and Treatment of Pharmaceutical Contaminants to Produce Clean Water for Better Health and Environmental. *Journal of Cleaner Production* **2023**, *387*, No. 135798.
- (90) Kumar, S.; Nehra, M.; Mehta, J.; Dilbaghi, N.; Marrazza, G.; Kaushik, A. Point-of-Care Strategies for Detection of Waterborne Pathogens. *Sensors* **2019**, *19* (20), 4476.
- (91) Wang, C.; Wang, P.; Chen, J.; Zhu, L.; Zhang, D.; Wan, Y.; Ai, S. Self-Powered Biosensing System Driven by Triboelectric Nanogenerator for Specific Detection of Gram-Positive Bacteria. *Nano Energy* **2022**, *93*, No. 106828.
- (92) Hara, T. O.; Singh, B. Electrochemical Biosensors for Detection of Pesticides and Heavy Metal Toxicants in Water: Recent Trends and Progress. *ACS EST Water* **2021**, *1* (3), 462–478.
- (93) Chee, G.-J.; Nomura, Y.; Ikebukuro, K.; Karube, I. Optical Fiber Biosensor for the Determination of Low Biochemical Oxygen Demand. *Biosens. Bioelectron.* **2000**, *15*, 371–376.
- (94) Bickman, S. R.; Campbell, K.; Elliott, C.; Murphy, C.; O’Kennedy, R.; Papst, P.; Lochhead, M. J. An Innovative Portable Biosensor System for the Rapid Detection of Freshwater Cyanobacterial Algal Bloom Toxins. *Environ. Sci. Technol.* **2018**, *52* (20), 11691–11698.
- (95) Orellana, G.; Villén, L.; Haigh, D.; Maneiro, E.; Marvá, F.; Costas, E. Microalgal Fiber-Optic Biosensors for Water Quality Monitoring. In *Third European Workshop on Optical Fibre Sensors*; SPIE, 2007; Vol. 6619, pp 347–350.
- (96) Fallah, H.; Asadishad, T.; Parsanasab, G. M.; Harun, S. W.; Mohammed, W. S.; Yasin, M. Optical Fiber Biosensor toward E-Coli Bacterial Detection on the Pollutant Water. *Engineering Journal* **2021**, *25* (12), 1–8.
- (97) Naessens, M.; Leclerc, J. C.; Tran-Minh, C. Fiber Optic Biosensor Using *Chlorella Vulgaris* for Determination of Toxic Compounds. *Ecotoxicology and Environmental Safety* **2000**, *46* (2), 181–185.
- (98) Lukyanenko, K. A.; Denisov, I. A.; Sorokin, V. V.; Yakimov, A. S.; Esimbekova, E. N.; Belobrov, P. I. Handheld Enzymatic Luminescent Biosensor for Rapid Detection of Heavy Metals in Water Samples. *Chemosensors* **2019**, *7* (1), 16.
- (99) Maleki, N.; Kashanian, S.; Maleki, E.; Nazari, M. A Novel Enzyme Based Biosensor for Catechol Detection in Water Samples Using Artificial Neural Network. *Biochemical Engineering Journal* **2017**, *128*, 1–11.
- (100) Coronado-Apodaca, K. G.; González-Meza, G. M.; Aguayo-Acosta, A.; Araújo, R. G.; Gonzalez-Gonzalez, R. B.; Oyervides-Muñoz, M. A.; Martínez-Ruiz, M.; Melchor-Martínez, E. M.; Barceló, D.; Parra-Saldivar, R.; Sosa-Hernández, J. E. Immobilized Enzyme-Based Novel Biosensing System for Recognition of Toxic Elements in the Aqueous Environment. *Top Catal* **2023**, *66* (9), 606–624.
- (101) Pasternak, G.; Greenman, J.; Ieropoulos, I. Self-Powered, Autonomous Biological Oxygen Demand Biosensor for Online Water Quality Monitoring. *Sens. Actuators, B* **2017**, *244*, 815–822.
- (102) Cunha, I.; Biltz, R.; Sales, M. G. F.; Vasconcelos, V. Aptamer-Based Biosensors to Detect Aquatic Phycotoxins and Cyanotoxins. *Sensors* **2018**, *18* (7), 2367.
- (103) Zhang, W.; Liu, Q. X.; Guo, Z. H.; Lin, J. S. Practical Application of Aptamer-Based Biosensors in Detection of Low Molecular Weight Pollutants in Water Sources. *Molecules* **2018**, *23* (2), 344.
- (104) Li, Y.; Zhu, Y.; Wang, C.; He, M.; Lin, Q. Selective Detection of Water Pollutants Using a Differential Aptamer-Based Graphene Biosensor. *Biosens. Bioelectron.* **2019**, *126*, 59–67.
- (105) Vishwakarma, A.; Lal, R.; Ramya, M. Aptamer-Based Approaches for the Detection of Waterborne Pathogens. *Int. Microbiol* **2021**, *24* (2), 125–140.
- (106) Soukarié, D.; Ecochard, V.; Salomé, L. DNA-Based Nano-biosensors for Monitoring of Water Quality. *International Journal of Hygiene and Environmental Health* **2020**, *226*, No. 113485.
- (107) Yildirim, N.; Long, F.; Gao, C.; He, M.; Shi, H.-C.; Gu, A. Z. Aptamer-Based Optical Biosensor For Rapid and Sensitive Detection of 17 β -Estradiol In Water Samples. *Environ. Sci. Technol.* **2012**, *46* (6), 3288–3294.
- (108) Fernández Blanco, A.; Moreno, Y.; García-Hernández, J.; Hernández, M. A Photonic Immunosensor Detection Method for

Viable and Non-Viable E. Coli in Water Samples. *Microorganisms* **2024**, *12* (7), 1328.

(109) Jögi, E.; Väling, I.; Rinken, T. Assessment of Bathing Water Quality with an E. Coli Immunosensor. *International Journal of Environmental Analytical Chemistry* **2022**, *102* (16), 4652–4663.

(110) Mauriz, E.; Calle, A.; Manclús, J. J.; Montoya, A.; Hildebrandt, A.; Barceló, D.; Lechuga, L. M. Optical Immunosensor for Fast and Sensitive Detection of DDT and Related Compounds in River Water Samples. *Biosens. Bioelectron.* **2007**, *22* (7), 1410–1418.

(111) Cimafronte, M.; Fulgione, A.; Gaglione, R.; Papaiani, M.; Capparelli, R.; Arciello, A.; Bolletti Censi, S.; Borriello, G.; Velotta, R.; Della Ventura, B. Screen Printed Based Impedimetric Immunosensor for Rapid Detection of Escherichia Coli in Drinking Water. *Sensors* **2020**, *20* (1), 274.

(112) Khatri, P.; Gupta, K. K.; Gupta, R. K. Raspberry Pi-Based Smart Sensing Platform for Drinking-Water Quality Monitoring System: A Python Framework Approach. *Drinking Water Engineering and Science* **2019**, *12* (1), 31–37.

(113) Wang, X.; Yang, W. Water Quality Monitoring and Evaluation Using Remote Sensing Techniques in China: A Systematic Review. *Ecosystem Health and Sustainability* **2019**, *5* (1), 47–56.

(114) Lu, Q.; Si, W.; Wei, L.; Li, Z.; Xia, Z.; Ye, S.; Xia, Y. Retrieval of Water Quality from UAV-Borne Hyperspectral Imagery: A Comparative Study of Machine Learning Algorithms. *Remote Sensing* **2021**, *13* (19), 3928.

(115) Niu, C.; Tan, K.; Jia, X.; Wang, X. Deep Learning Based Regression for Optically Inactive Inland Water Quality Parameter Estimation Using Airborne Hyperspectral Imagery. *Environ. Pollut.* **2021**, *286*, No. 117534.

(116) Canciu, A.; Tertis, M.; Hosu, O.; Cernat, A.; Cristea, C.; Graur, F. Modern Analytical Techniques for Detection of Bacteria in Surface and Wastewaters. *Sustainability* **2021**, *13* (13), 7229.

(117) Su, X.; Sutarlie, L.; Loh, X. J. Sensors, Biosensors, and Analytical Technologies for Aquaculture Water Quality. *Research* **2020**, *2020*, DOI: 10.34133/2020/8272705.

(118) Du, R.; Yang, D.; Yin, X. Rapid Detection of Three Common Bacteria Based on Fluorescence Spectroscopy. *Sensors (Basel)* **2022**, *22* (3), 1168.

(119) Paz, E. F. M.; Tobias, M.; Escobar, E.; Raskin, L.; Roberts, E. F. S.; Wigginton, K. R.; Kerkez, B. Wireless Sensors for Measuring Drinking Water Quality in Building Plumbing: Deployments and Insights from Continuous and Intermittent Water Supply Systems. *ACS ES&T Engineering* **2022**, *2*, 423.

(120) Martinez Paz, E. F.; Tobias, M.; Escobar, E.; Raskin, L.; Roberts, E. F. S.; Wigginton, K. R.; Kerkez, B. Wireless Sensors for Measuring Drinking Water Quality in Building Plumbing: Deployments and Insights from Continuous and Intermittent Water Supply Systems. *ACS EST Eng.* **2022**, *2* (3), 423–433.

(121) Gong, L.; Martinez, O.; Mesquita, P.; Kurtz, K.; Xu, Y.; Lin, Y. A Microfluidic Approach for Label-Free Identification of Small-Sized Microplastics in Seawater. *Sci. Rep.* **2023**, *13* (1), No. 11011.

(122) Ahuja, S.; Kumar, M. S.; Nandeshwar, R.; Kondabagil, K.; Tallur, S. Longer Amplicons Provide Better Sensitivity for Electrochemical Sensing of Viral Nucleic Acid in Water Samples Using PCB Electrodes. *Sci. Rep.* **2022**, *12*, DOI: 10.1038/s41598-022-12818-w.

(123) Jain, S.; Dilbaghi, N.; Kumar Singhal, N.; Kaushik, A.; Kim, K.-H.; Kumar, S. Carbon Quantum Dots@metal–Organic Framework Based Catalytic Nucleic Acid Fluorescent System for Highly Sensitive and Selective Detection of Pb²⁺ in Aqueous Solutions. *Chemical Engineering Journal* **2023**, *457*, No. 141375.

(124) Lowe, M.; Qin, R.; Mao, X. A Review on Machine Learning, Artificial Intelligence, and Smart Technology in Water Treatment and Monitoring. *Water* **2022**, *14* (9), 1384.

(125) Chowdury, M. S. U.; Emran, T. B.; Ghosh, S.; Pathak, A.; Alam, M. M.; Absar, N.; Andersson, K.; Hossain, M. S. IoT Based Real-Time River Water Quality Monitoring System. *Procedia Computer Science* **2019**, *155*, 161–168.

(126) Jan, F.; Min-Allah, N.; Düşteğör, D. IoT Based Smart Water Quality Monitoring: Recent Techniques, Trends and Challenges for Domestic Applications. *Water* **2021**, *13* (13), 1729.

(127) Liu, P.; Wang, J.; Sangaiah, A. K.; Xie, Y.; Yin, X. Analysis and Prediction of Water Quality Using LSTM Deep Neural Networks in IoT Environment. *Sustainability* **2019**, *11* (7), 2058.

(128) Pasika, S.; Gandla, S. T. Smart Water Quality Monitoring System with Cost-Effective Using IoT. *Heliyon* **2020**, *6* (7), No. e04096.

(129) Ramadhan, A. J. Smart Water-Quality Monitoring System Based on Enabled Real-Time Internet of Things. *J. Eng. Sci. Technol.* **2020**, *15*, 3514–3527.

(130) Laad, M.; Ghule, B. Removal of Toxic Contaminants from Drinking Water Using Biosensors: A Systematic Review. *Groundwater for Sustainable Development* **2023**, *20*, No. 100888.

(131) Jagtap, S.; Skouteris, G.; Choudhari, V.; Rahimifard, S. Improving Water Efficiency in the Beverage Industry With the Internet of Things. In *Implementing Data Analytics and Architectures for Next Generation Wireless Communications*; IGI Global, 2022; pp 18–26.

(132) Ahad, M. A.; Paiva, S.; Tripathi, G.; Feroz, N. Enabling Technologies and Sustainable Smart Cities. *Sustainable Cities and Society* **2020**, *61*, No. 102301.

(133) Ataei Kachouei, M.; Kaushik, A.; Ali, Md. A. Internet of Things-Enabled Food and Plant Sensors to Empower Sustainability. *Advanced Intelligent Systems* **2023**, *5*, No. 2300321.

(134) Kasinathan, P.; Pugazhendhi, R.; Elavarasan, R. M.; Ramchandaramurthy, V. K.; Ramanathan, V.; Subramanian, S.; Kumar, S.; Nandhagopal, K.; Raghavan, R. R. V.; Rangasamy, S.; Devendiran, R.; Alsharif, M. H. Realization of Sustainable Development Goals with Disruptive Technologies by Integrating Industry 5.0, Society 5.0, Smart Cities and Villages. *Sustainability* **2022**, *14* (22), 15258.

(135) Mishra, P. *Sustainable Smart Cities: Enabling Technologies, Energy Trends and Potential Applications*; Springer Nature, 2023.

(136) Tabaa, M.; Monteiro, F.; Bensag, H.; Dandache, A. Green Industrial Internet of Things from a Smart Industry Perspectives. *Energy Reports* **2020**, *6*, 430–446.

(137) Kakhi, K.; Alizadehsani, R.; Kabir, H. M. D.; Khosravi, A.; Nahavandi, S.; Acharya, U. R. The Internet of Medical Things and Artificial Intelligence: Trends, Challenges, and Opportunities. *Biocybernetics and Biomedical Engineering* **2022**, *42* (3), 749–771.

(138) Mustafa, H. M.; Mustapha, A.; Hayder, G.; Salisu, A. Applications of IoT and Artificial Intelligence in Water Quality Monitoring and Prediction: A Review. *2021 6th International Conference on Inventive Computation Technologies (ICICT)*; 2021, pp 968–975.

(139) Al-Dhaen, F.; Hou, J.; Rana, N. P.; Weerakkody, V. Advancing the Understanding of the Role of Responsible AI in the Continued Use of IoMT in Healthcare. *Inf Syst.* **2023**, *25*, 2159–2178.

(140) Manickam, P.; Mariappan, S. A.; Murugesan, S. M.; Hansda, S.; Kaushik, A.; Shinde, R.; Thipperudraswamy, S. P. Artificial Intelligence (AI) and Internet of Medical Things (IoMT) Assisted Biomedical Systems for Intelligent Healthcare. *Biosensors* **2022**, *12* (8), 562.

(141) Chakraborty, U.; Kaushik, A.; Chaudhary, G. R.; Mishra, Y. K. Emerging Nano-Enabled Gas Sensor for Environmental Monitoring – Perspectives and Open Challenges. *Current Opinion in Environmental Science & Health* **2024**, *37*, No. 100532.

(142) Peng, J.; Cai, K.; Jin, X. High Concurrency Massive Data Collection Algorithm for IoMT Applications. *Computer Communications* **2020**, *157*, 402–409.

(143) Bhardwaj, A.; Dagar, V.; Khan, M.; Aggarwal, A.; Alvarado, R.; Kumar, M.; Irfan, M.; Proshad, R. Smart IoT and Machine Learning-Based Framework for Water Quality Assessment and Device Component Monitoring. *Environmental Science and Pollution Research* **2022**, *29*, 46018.

(144) Song, Y.; Shin, H.; Koo, S.; Baek, S.; Seo, J.; Kang, H.; Kim, Y. Internet of Maritime Things Platform for Remote Marine Water

- Quality Monitoring. *IEEE Internet of Things Journal* **2022**, *9* (16), 14355–14365.
- (145) Mushtaq, M. S.; Mushtaq, Y.; Raza, M. Q.; Hussain, S. A. 4 - Performance Enhancement of IoMT Using Artificial Intelligence Algorithms. In *Security and Privacy Issues in Internet of Medical Things*; Buyya, R., Tariq, M. I., Balas, V. E., Wang, G., Prodan, R., Eds.; Academic Press, 2023; pp 61–81.
- (146) Mustapha, U. F.; Alhassan, A.-W.; Jiang, D.-N.; Li, G.-L. Sustainable Aquaculture Development: A Review on the Roles of Cloud Computing, Internet of Things and Artificial Intelligence (CIA). *Reviews in Aquaculture* **2021**, *13* (4), 2076–2091.
- (147) Paepae, T.; Bokoro, P. N.; Kyamakya, K. From Fully Physical to Virtual Sensing for Water Quality Assessment: A Comprehensive Review of the Relevant State-of-the-Art. *Sensors (Basel)* **2021**, *21* (21), 6971.
- (148) Yi, J.; Wisuthiphaet, N.; Raja, P.; Nitin, N.; Earles, J. M. AI-Enabled Biosensing for Rapid Pathogen Detection: From Liquid Food to Agricultural Water. *Water Res.* **2023**, *242*, No. 120258.
- (149) Yang, Y.; Xu, F.; Chen, J.; Tao, C.; Li, Y.; Chen, Q.; Tang, S.; Lee, H. K.; Shen, W. Artificial Intelligence-Assisted Smartphone-Based Sensing for Bioanalytical Applications: A Review. *Biosens. Bioelectron.* **2023**, *229*, No. 115233.
- (150) Abdeldayem, O. M.; Dabbish, A. M.; Habashy, M. M.; Mostafa, M. K.; Elhefnawy, M.; Amin, L.; Al-Sakkari, E. G.; Ragab, A.; Rene, E. R. Viral Outbreaks Detection and Surveillance Using Wastewater-Based Epidemiology, Viral Air Sampling, and Machine Learning Techniques: A Comprehensive Review and Outlook. *Science of The Total Environment* **2022**, *803*, No. 149834.
- (151) Shanmuga Sundari, M.; Samyuktha, P.; Kranthi, A.; Das, S. Evaluating Performance on Covid-19 Tweet Sentiment Analysis Outbreak Using Support Vector Machine. In *Smart Intelligent Computing and Applications*; Bhateja, V., Satapathy, S. C., Travieso-Gonzalez, C. M., Adilakshmi, T., Eds.; Springer Nature: Singapore, 2022; pp 151–159; Vol. 1.
- (152) Das, S. S. M. S.; Mazumdar, H. Predictive Accuracy Analysis for Corona Virus Using Residual U-Net in Radiological Data. *2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)*; 2024, Vol. 2, pp 1–6.
- (153) Gopi, A. P.; Naik, K. J. A Model for Analysis of IoT Based Aquarium Water Quality Data Using CNN Model *2021 International Conference on Decision Aid Sciences and Application (DASA)*; 2021, pp 976–980.
- (154) Maity, A.; Pu, H.; Sui, X.; Chang, J.; Bottum, K. J.; Jin, B.; Zhou, G.; Wang, Y.; Lu, G.; Chen, J. Scalable Graphene Sensor Array for Real-Time Toxins Monitoring in Flowing Water. *Nat. Commun.* **2023**, *14* (1), 4184.
- (155) Rustam, F.; Ishaq, A.; Kokab, S. T.; de la Torre Diez, I.; Mazón, J. L. V.; Rodríguez, C. L.; Ashraf, I. An Artificial Neural Network Model for Water Quality and Water Consumption Prediction. *Water* **2022**, *14* (21), 3359.
- (156) Aldhyani, T. H. H.; Al-Yaari, M.; Alkahtani, H.; Maashi, M. Water Quality Prediction Using Artificial Intelligence Algorithms. *Applied Bionics and Biomechanics* **2020**, *2020*, No. e6659314.
- (157) Zhu, M.; Wang, J.; Yang, X.; Zhang, Y.; Zhang, L.; Ren, H.; Wu, B.; Ye, L. A Review of the Application of Machine Learning in Water Quality Evaluation. *Eco-Environment & Health* **2022**, *1* (2), 107–116.
- (158) Nasir, N.; Kansal, A.; Alshaltone, O.; Barneih, F.; Sameer, M.; Shanableh, A.; Al-Shamma'a, A. Water Quality Classification Using Machine Learning Algorithms. *Journal of Water Process Engineering* **2022**, *48*, No. 102920.
- (159) Shyu, H.-Y.; Castro, C. J.; Bair, R. A.; Lu, Q.; Yeh, D. H. Development of a Soft Sensor Using Machine Learning Algorithms for Predicting the Water Quality of an Onsite Wastewater Treatment System. *ACS Environ. Au* **2023**, *3* (5), 308–318.
- (160) Djerioui, M.; Bouamar, M.; Ladjal, M.; Zerguine, A. Chlorine Soft Sensor Based on Extreme Learning Machine for Water Quality Monitoring. *Arab J. Sci. Eng.* **2019**, *44* (3), 2033–2044.
- (161) Guo, J.; Lee, J. H. W. Development of Predictive Models for “Very Poor” Beach Water Quality Gradings Using Class-Imbalance Learning. *Environ. Sci. Technol.* **2021**, *55* (21), 14990–15000.
- (162) Hussain, M.; Srividya Devi, P.; Rao, C. R. V.; Raju, V. V. R.; Kulkarni, S. Application of Multiple Linear Regression and Artificial Neural Networks in River Water Quality Modelling to Predict Dissolved Oxygen in Rivers: A Case Study of Krishna River in India. *AIP Conf. Proc.* **2023**, *2754* (1), No. 120002.
- (163) Pattanayak, A. S.; Pattnaik, B. S.; Udgata, S. K.; Panda, A. K. Development of Chemical Oxygen on Demand (COD) Soft Sensor Using Edge Intelligence. *IEEE Sensors Journal* **2020**, *20* (24), 14892–14902.
- (164) Guselnikova, O.; Trelin, A.; Kang, Y.; Postnikov, P.; Kobashi, M.; Suzuki, A.; Shrestha, L. K.; Henzie, J.; Yamauchi, Y. Pretreatment-Free SERS Sensing of Microplastics Using a Self-Attention-Based Neural Network on Hierarchically Porous Ag Foams. *Nat. Commun.* **2024**, *15* (1), 4351.
- (165) Zhou, J.; Wu, Z.; Jin, C.; Zhang, J. X. J. Machine Learning Assisted Dual-Functional Nanophotonic Sensor for Organic Pollutant Detection and Degradation in Water. *npj Clean Water* **2024**, *7* (1), 1–11.
- (166) Sudhakar, M. Artificial Intelligence Applications in Water Treatment and Water Resource Assessment: Challenges, Innovations, and Future Directions. *Intelligent Engineering Applications and Applied Sciences for Sustainability*; IGI Global, 2023; pp 248–269.
- (167) Haggerty, R.; Sun, J.; Yu, H.; Li, Y. Application of Machine Learning in Groundwater Quality Modeling - A Comprehensive Review. *Water Res.* **2023**, *233*, No. 119745.
- (168) Zaroni, M. G.; Majone, B.; Bellin, A. A Catchment-Scale Model of River Water Quality by Machine Learning. *Science of The Total Environment* **2022**, *838*, No. 156377.
- (169) Tianan, D.; Chau, K.; Duan, H.-F. Machine Learning Based Marine Water Quality Prediction for Coastal Hydro-Environment Management. *Journal of Environmental Management* **2021**, *284*, No. 112051.
- (170) Nieves, V.; Radin, C.; Camps-Valls, G. Predicting Regional Coastal Sea Level Changes with Machine Learning. *Sci. Rep* **2021**, *11* (1), 7650.
- (171) Ahlawat, W.; Dilbaghi, N.; Kumar, R.; Singhal, N. K.; Kaushik, A.; Kumar, S. Adsorption of Harmful Dyes and Antimicrobial Studies Utilizing Recyclable ZnO, Its Composites with Conventionally Used Activated Carbon, and Waste Orange Peel as a Greener Approach. *Journal of Environmental Chemical Engineering* **2023**, *11* (3), No. 110268.
- (172) Preeti, K.; Kumar, A.; Jain, N.; Kaushik, A.; Mishra, Y. K.; Sharma, S. K. Tailored ZnO Nanostructures for Efficient Sensing of Toxic Metallic Ions of Drainage Systems. *Materials Today Sustainability* **2023**, *24*, No. 100515.
- (173) Singh, M.; Kapur, A.; Chauhan, M.; Kaur, G.; Kaushik, A.; Chaudhary, G. R. Green Arginine Capped Hafnium Oxide Nanoparticles, a Computationally Designed Framework, for Electrochemical Sensing of Mercury (II) Ion. *Chemical Engineering Journal* **2023**, *469*, No. 144075.
- (174) Zhao, F.; Liu, M.; Wang, K.; Wang, T.; Jiang, X. A Soft Measurement Approach of Wastewater Treatment Process by Lion Swarm Optimizer-Based Extreme Learning Machine. *Measurement* **2021**, *179*, No. 109322.
- (175) Chen, Y.; Song, L.; Liu, Y.; Yang, L.; Li, D. A Review of the Artificial Neural Network Models for Water Quality Prediction. *Applied Sciences* **2020**, *10* (17), 5776.
- (176) Singha, S.; Pasupuleti, S.; Singha, S. S.; Singh, R.; Kumar, S. Prediction of Groundwater Quality Using Efficient Machine Learning Technique. *Chemosphere* **2021**, *276*, No. 130265.
- (177) Karthikeyan, M.; Vijayachitra, S.; Pyngkodi, M.; Arunkumar, S.; Kaviya, K. N.; Madhumitha, R. Study on Water Quality and Wastewater Treatment Using IOT. *2022 6th International Conference on Computing Methodologies and Communication (ICCMC)*; 2022, pp 431–438.

- (178) Bhagat, S. K.; Pyrgaki, K.; Salih, S. Q.; Tiyyasha, T.; Beyaztas, U.; Shahid, S.; Yaseen, Z. M. Prediction of Copper Ions Adsorption by Attapulgite Adsorbent Using Tuned-Artificial Intelligence Model. *Chemosphere* **2021**, 276, No. 130162.
- (179) Yan, T.; Shen, S.-L.; Zhou, A. Indices and Models of Surface Water Quality Assessment: Review and Perspectives. *Environ. Pollut.* **2022**, 308, No. 119611.
- (180) Thorslund, J.; van Vliet, M. T. H. A Global Dataset of Surface Water and Groundwater Salinity Measurements from 1980–2019. *Scientific Data* **2020**, 7 (1), 231.
- (181) Rai, H. M.; Atik-Ur-Rehman; Pal, A.; Mishra, S.; Shukla, K. K. Use of Internet of Things in the Context of Execution of Smart City Applications: A Review. *Discov Internet Things* **2023**, 3 (1), 8.
- (182) Molinara, M.; Bria, A.; De Vito, S.; Marrocco, C. Artificial Intelligence for Distributed Smart Systems. *Pattern Recognition Letters* **2021**, 142, 48–50.
- (183) Zhou, Y. Real-Time Probabilistic Forecasting of River Water Quality under Data Missing Situation: Deep Learning plus Post-Processing Techniques. *Journal of Hydrology* **2020**, 589, No. 125164.
- (184) Vasudevan, S. K.; Baskaran, B. An Improved Real-Time Water Quality Monitoring Embedded System with IoT on Unmanned Surface Vehicle. *Ecological Informatics* **2021**, 65, No. 101421.
- (185) Hagar, J.; Murray, R.; Haxton, T.; Hall, J.; McKenna, S. Using the CANARY Event Detection Software to Enhance Security and Improve Water Quality. *World Environmental and Water Resources Congress 2013: Showcasing the Future*; 2013; pp 989–1003.
- (186) Leow, A.; Burkhardt, J.; Platten, W. E., III; Zimmerman, B.; Brinkman, N. E.; Turner, A.; Murray, R.; Sorial, G.; Garland, J. Application of the CANARY Event Detection Software for Real-Time Performance Monitoring of Decentralized Water Reuse Systems. *Environ. Sci.: Water Res. Technol.* **2017**, 3 (2), 224–234.
- (187) Nafsin, N.; Li, J. Using CANARY Event Detection Software for Water Quality Analysis in the Milwaukee River. *Journal of Hydro-environment Research* **2021**, 38, 117–128.
- (188) Clean Water AI for drinking water safety - Microsoft AI Lab. <https://www.microsoft.com/en-us/ai/ai-lab-clean-water> (accessed 2023-09-26).
- (189) Leveraging partnerships, technology, and design to enhance water quality through the inaugural Aquality app. IBM Blog. <https://www.ibm.com/blog/announcement/leveraging-partnerships-technology-and-design-to-enhance-water-quality-through-the-inaugural-aquality-app/> (accessed 2023-09-15).
- (190) Aqueduct. World Resources Institute. <https://www.wri.org/aqueduct> (accessed 2023-09-26).
- (191) A 'Smart' step towards clean drinking water in rural India - Impact stories. Tata Trusts. <https://www.tatatrusts.org/our-stories/article/a-smart-step-towards-clean-drinking-water-in-rural-india> (accessed 2023-09-15).
- (192) Swimming. <https://www.waternet.nl/en/our-water/safe-swimming/> (accessed 2023-09-26).
- (193) Peters, S.; Ouboter, M.; van der Lugt, K.; Koop, S.; van Leeuwen, K. Retrospective Analysis of Water Management in Amsterdam, The Netherlands. *Water* **2021**, 13 (8), 1099.
- (194) Monitoring. Chesapeake Bay. <https://www.chesapeakebay.net/what/programs/monitoring> (accessed 2023-09-26).
- (195) Turning the Tide: The Digital Future of Water Management. <https://www.autodesk.com/design-make/articles/water-management> (accessed 2023-09-26).
- (196) Barajas, M. A.; Murphy, M. P.; Lasseter, L. C.; Sunny, G. I.; Mazumdar, H.; Gohel, H. A.; Emerson, H. P.; Kaplan, D. I. Seasonal Trend Assessment for Groundwater Contamination Detection and Monitoring Using ARIMA Model. *2023 IEEE 2nd International Conference on AI in Cybersecurity (ICAIC)* **2023**, 1–7.
- (197) Murphy, M. P.; Mazumdar, H.; Gohel, H. A.; Emerson, H. P.; Kaplan, D. I. Long Short-Term Memory Networks for Monitoring Groundwater Contamination at the Hanford Site. *2023 IEEE 2nd International Conference on AI in Cybersecurity (ICAIC)* **2023**, 1–5.
- (198) Hanf, R. W.; Kelly, L. M. An Assessment of Drinking-Water Supplies on the Hanford Site: An Evaluation Conducted at a Federal Nuclear Facility in Southeastern Washington State. *J. Environ. Health* **2005**, 67 (7), 44–49.
- (199) Guay, C. Water and technology: A sustainable future. IBM Blogs - Canada. <https://www.ibm.com/blogs/ibm-canada/2021/03/water-and-technology-a-sustainable-future/> (accessed 2023-09-26).
- (200) Elkiran, G.; Nourani, V.; Abba, S. I.; Abdullahi, J. Artificial Intelligence-Based Approaches for Multi-Station Modelling of Dissolve Oxygen in River. *Global Journal of Environmental Science and Management* **2018**, 4 (4), 439–450.
- (201) Nourani, V.; Asghari, P.; Sharghi, E. Artificial Intelligence Based Ensemble Modeling of Wastewater Treatment Plant Using Jittered Data. *Journal of Cleaner Production* **2021**, 291, No. 125772.
- (202) Saboe, D.; Ghasemi, H.; Gao, M. M.; Samardzic, M.; Hristovski, K. D.; Boscovic, D.; Burge, S. R.; Burge, R. G.; Hoffman, D. A. Real-Time Monitoring and Prediction of Water Quality Parameters and Algae Concentrations Using Microbial Potentiometric Sensor Signals and Machine Learning Tools. *Science of The Total Environment* **2021**, 764, No. 142876.
- (203) Batur, E.; Maktav, D. Assessment of Surface Water Quality by Using Satellite Images Fusion Based on PCA Method in the Lake Gala, Turkey. *IEEE Transactions on Geoscience and Remote Sensing* **2019**, 57 (5), 2983–2989.
- (204) Onag, G. IoT helps bring clean water to Indian villages. FutureIoT. <https://futureiot.tech/iot-helps-bring-clean-water-to-indian-villages/> (accessed 2023-09-15).
- (205) Yan, J.; Xu, Z.; Yu, Y.; Xu, H.; Gao, K. Application of a Hybrid Optimized BP Network Model to Estimate Water Quality Parameters of Beihai Lake in Beijing. *Applied Sciences* **2019**, 9 (9), 1863.
- (206) Wei, Y.; Jiao, Y.; An, D.; Li, D.; Li, W.; Wei, Q. Review of Dissolved Oxygen Detection Technology: From Laboratory Analysis to Online Intelligent Detection. *Sensors* **2019**, 19 (18), 3995.
- (207) BridgeThings. Case study: Monitoring water quality parameters across the Ganga River as part of the NMCG Project. BridgeThings. <https://www.bridgethings.com/case-study-water-quality-monitoring-across-ganga-river/> (accessed 2023-09-08).
- (208) Singh, S.; Rai, S.; Singh, P.; Mishra, V. K. Real-Time Water Quality Monitoring of River Ganga (India) Using Internet of Things. *Ecological Informatics* **2022**, 71, No. 101770.
- (209) Post, C.; Brülisauer, S.; Waldschläger, K.; Hug, W.; Grüneis, L.; Heyden, N.; Schmor, S.; Förderer, A.; Reid, R.; Reid, M.; Bhartia, R.; Nguyen, Q.; Schüttrumpf, H.; Amann, F. Application of Laser-Induced, Deep UV Raman Spectroscopy and Artificial Intelligence in Real-Time Environmental Monitoring—Solutions and First Results. *Sensors* **2021**, 21 (11), 3911.
- (210) Noumani, A.; Verma, D.; Kaushik, A.; Khosla, A.; Solanki, P. R. Electrochemical Microplastic Detection Using Chitosan-Magnesium Oxide Nanosheet. *Environmental Research* **2024**, 252, No. 118894.
- (211) Solving Water Quality Monitoring Challenges in the Oil and Gas Industry - KETOS. <https://ketos.com/solving-water-quality-monitoring-challenges-in-the-oil-and-gas-industry> (accessed 2023-09-26).
- (212) Adeleke, I. A.; Nwulu, N. I.; Ogbolumani, O. A. A Hybrid Machine Learning and Embedded IoT-Based Water Quality Monitoring System. *Internet of Things* **2023**, 22, No. 100774.
- (213) Ajakwe, S. O.; Ajakwe, I. U.; Jun, T.; Kim, D.-S.; Lee, J.-M. CIS-WQMS: Connected Intelligence Smart Water Quality Monitoring Scheme. *Internet of Things* **2023**, 23, No. 100800.
- (214) Elhadad, A.; Alanazi, F.; Taloba, A. I.; Abozeid, A. Fog Computing Service in the Healthcare Monitoring System for Managing the Real-Time Notification. *Journal of Healthcare Engineering* **2022**, 2022, No. e5337733.
- (215) Mohamed, N.; Al-Jaroodi, J.; Jawhar, I. Towards Fault Tolerant Fog Computing for IoT-Based Smart City Applications. *2019 IEEE 9th Annual Computing and Communication Workshop and Conference (CCWC)* **2019**, 0752–0757.
- (216) Lin, Y.-P.; Mukhtar, H.; Huang, K.-T.; Petway, J. R.; Lin, C.-M.; Chou, C.-F.; Liao, S.-W. Real-Time Identification of Irrigation Water Pollution Sources and Pathways with a Wireless Sensor Network and Blockchain Framework. *Sensors* **2020**, 20 (13), 3634.

- (217) Alharbi, N.; Althagafi, A.; Alshomrani, O.; Almotiry, A.; Alhazmi, S. A Blockchain Based Secure IoT Solution for Water Quality Management. *2021 International Congress of Advanced Technology and Engineering (ICOTEN)* **2021**, 1–8.
- (218) Drăgulescu, A.-M.; Constantin, F.; Orza, O.; Bosoc, S.; Streche, R.; Negoita, A.; Osiac, F.; Balaceanu, C.; Suciuc, G. Smart Watering System Security Technologies Using Blockchain. *2021 13th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)* **2021**, 1–4.
- (219) Zeng, H.; Dhiman, G.; Sharma, A.; Sharma, A.; Tselykh, A. An IoT and Blockchain-Based Approach for the Smart Water Management System in Agriculture. *Expert Systems* **2023**, 40 (4), No. e12892.
- (220) Lakshmikantha, V.; Hiriyannagowda, A.; Manjunath, A.; Patted, A.; Basavaiah, J.; Anthony, A. A. IoT Based Smart Water Quality Monitoring System. *Global Transitions Proceedings* **2021**, 2 (2), 181–186.
- (221) Park, J.; Ahn, J.; Kim, J.; Yoon, Y.; Park, J. Prediction and Interpretation of Water Quality Recovery after a Disturbance in a Water Treatment System Using Artificial Intelligence. *Water* **2022**, 14 (15), 2423.
- (222) Park, J.; Lee, W. H.; Kim, K. T.; Park, C. Y.; Lee, S.; Heo, T.-Y. Interpretation of Ensemble Learning to Predict Water Quality Using Explainable Artificial Intelligence. *Science of The Total Environment* **2022**, 832, No. 155070.
- (223) Rodríguez-López, L.; Bustos Usta, D.; Bravo Alvarez, L.; Duran-Llaser, I.; Lami, A.; Martínez-Retureta, R.; Urrutia, R. Machine Learning Algorithms for the Estimation of Water Quality Parameters in Lake Llanquihue in Southern Chile. *Water* **2023**, 15 (11), 1994.
- (224) Wu, X.; Zhang, Q.; Wen, F.; Qi, Y. A Water Quality Prediction Model Based on Multi-Task Deep Learning: A Case Study of the Yellow River, China. *Water* **2022**, 14 (21), 3408.
- (225) Miry, A.; Aramice, G. Water Monitoring and Analytic Based ThingSpeak. *International Journal of Electrical and Computer Engineering* **2020**, 10, 3588–3595.
- (226) Mazumdar, H.; Murphy, M. P.; Bhatkande, S.; Emerson, H. P.; Kaplan, D. I.; Gohel, H. A. Optimized Machine Learning Model for Predicting Groundwater Contamination. *2022 IEEE MetroCon* **2022**, 1–3.
- (227) Gambin, Á. F.; Angelats, E.; González, J. S.; Miozzo, M.; Dini, P. Sustainable Marine Ecosystems: Deep Learning for Water Quality Assessment and Forecasting. *IEEE Access* **2021**, 9, 121344–121365.
- (228) Hemdan, E. E.-D.; Essa, Y. M.; Shouman, M.; El-Sayed, A.; Moustafa, A. N. An Efficient IoT Based Smart Water Quality Monitoring System. *Multimed Tools Appl.* **2023**, 82 (19), 28827–28851.
- (229) Lafont, M.; Dupont, S.; Cousin, P.; Vallauri, A.; Dupont, C. Back to the Future: IoT to Improve Aquaculture: Real-Time Monitoring and Algorithmic Prediction of Water Parameters for Aquaculture Needs. *2019 Global IoT Summit (GloTS)* **2019**, 1–6.
- (230) Vo, D. T.; Nguyen, X. P.; Nguyen, T. D.; Hidayat, R.; Huynh, T. T.; Nguyen, D. T. A Review on the Internet of Thing (IoT) Technologies in Controlling Ocean Environment. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* **2021**, 0 (0), 1–19.
- (231) Manjakkal, L.; Mitra, S.; Petillot, Y. R.; Shutler, J.; Scott, E. M.; Willander, M.; Dahiya, R. Connected Sensors, Innovative Sensor Deployment, and Intelligent Data Analysis for Online Water Quality Monitoring. *IEEE Internet of Things Journal* **2021**, 8 (18), 13805–13824.
- (232) Chhipi-Shrestha, G.; Mian, H. R.; Mohammadiun, S.; Rodriguez, M.; Hewage, K.; Sadiq, R. Digital Water: Artificial Intelligence and Soft Computing Applications for Drinking Water Quality Assessment. *Clean Techn Environ. Policy* **2023**, 25 (5), 1409–1438.
- (233) Jayaraman, P.; Nagarajan, K. K.; Partheeban, P.; Krishnamurthy, V. Critical Review on Water Quality Analysis Using IoT and Machine Learning Models. *International Journal of Information Management Data Insights* **2024**, 4 (1), No. 100210.
- (234) Gunasekaran, K.; Boopathi, S. Artificial Intelligence in Water Treatments and Water Resource Assessments. In *Artificial Intelligence Applications in Water Treatment and Water Resource Management*; IGI Global, 2023; pp 71–98.
- (235) Mazumdar, H.; Khondakar, K. R.; Das, S.; Kaushik, A. Aspects of 6th Generation Sensing Technology: From Sensing to Sense. *Front. Nanotechnol.* **2024**, 6, DOI: 10.3389/fnano.2024.1434014.
- (236) D'Amico, G.; L'Abbate, P.; Liao, W.; Yigitcanlar, T.; Ioppolo, G. Understanding Sensor Cities: Insights from Technology Giant Company Driven Smart Urbanism Practices. *Sensors* **2020**, 20 (16), 4391.
- (237) Tyagi, N.; Sharma, G.; Kumar, D.; Pratap Neelratan, P.; Sharma, D.; Khanuja, M.; Singh, M. K.; Singh, V.; Kaushik, A.; Sharma, S. K. 2D-MXenes to Tackle Wastewater: From Purification to SERS-Based Sensing. *Coord. Chem. Rev.* **2023**, 496, No. 215394.
- (238) Uthappa, U. T.; Nehra, M.; Kumar, R.; Dilbaghi, N.; Marrazza, G.; Kaushik, A.; Kumar, S. Trends and Prospects of 2-D Tungsten Disulphide (WS₂) Hybrid Nanosystems for Environmental and Biomedical Applications. *Adv. Colloid Interface Sci.* **2023**, 322, No. 103024.
- (239) Das, S.; Mazumdar, H.; Khondakar, K. R.; Kaushik, A. Machine Learning Assisted Enhancement in a Two-Dimensional Material's Sensing Performance. *ACS Appl. Nano Mater.* **2024**, 7 (12), 13893–13918.